

HEDDY

Desk sculpture | build manual

1,446 pieces

217 mm tall | 148 steps

Think it through with Heddy.

heddy.app



How this build works

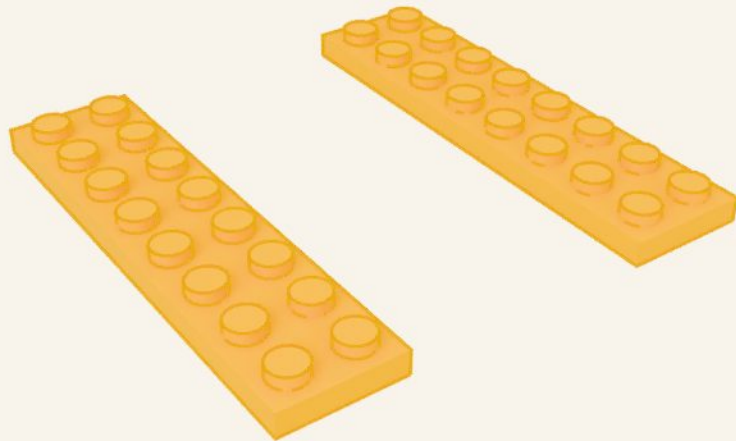
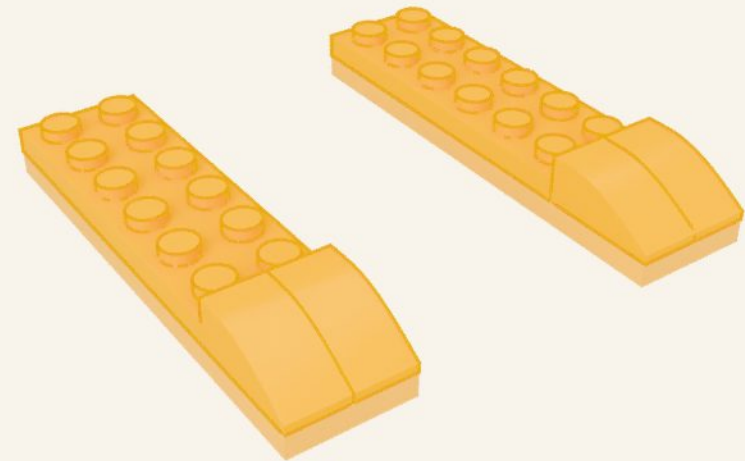
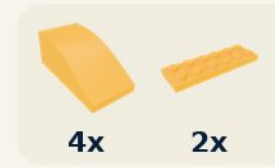
Heddy is built like a real sculpture: a hollow shell of plates and bricks, one layer at a time. Her face is built sideways (studs forward) and clicked onto bricks with studs on the side. A hidden turntable lets her upper body turn on her feet. Grey inside = structure you will not see. Each step shows the new pieces in full colour; pieces from earlier steps are shown lighter.

Build stages

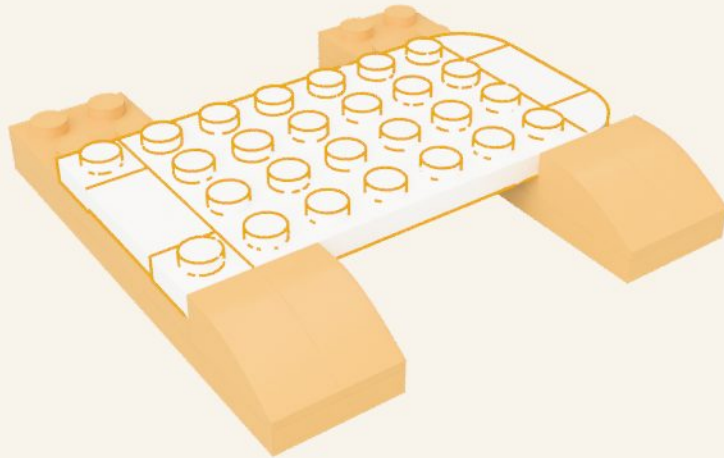
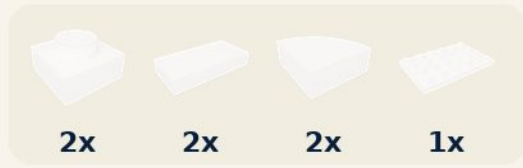
1	Feet	steps 1-2
2	Lower body	steps 3-9
3	Internal core & swivel	steps 10-18
4	Torso & wings	steps 19-59
5	Head structure	steps 60-133
6	Facial geometry	steps 134-138
7	Eyes	steps 139-144
8	Beak	steps 145-146
9	Finishing details	steps 147-148

Colours

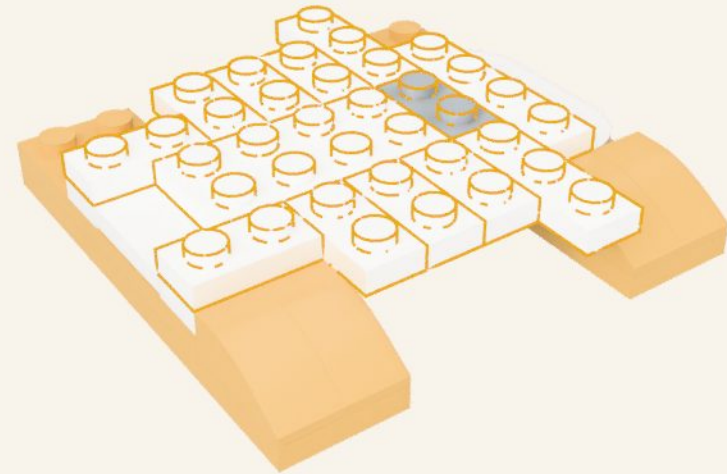
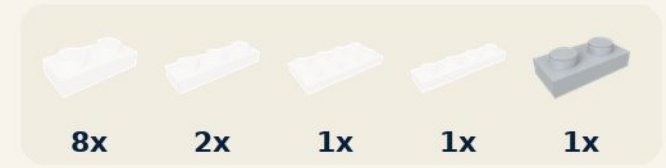
	White	951
	Medium Stone Grey	310
	Light Stone Grey	152
	Flame Yellowish Orange	18
	Earth Blue	12
	Dark Green	2
	Dark Stone Grey	1

1**2**

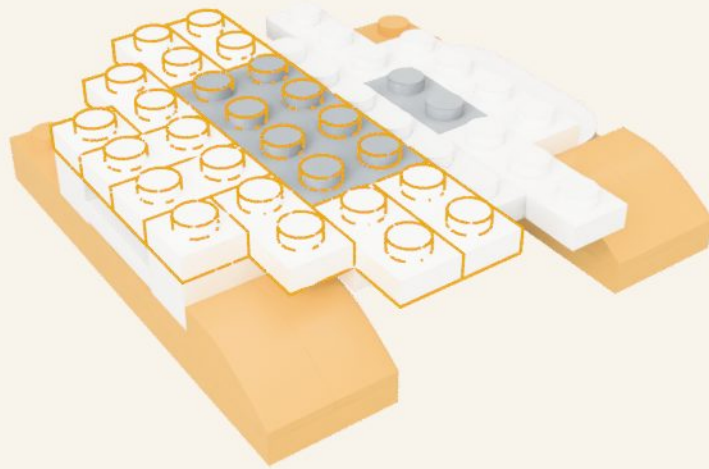
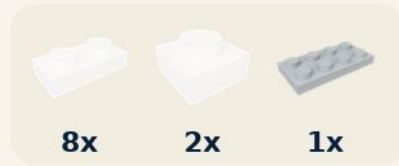
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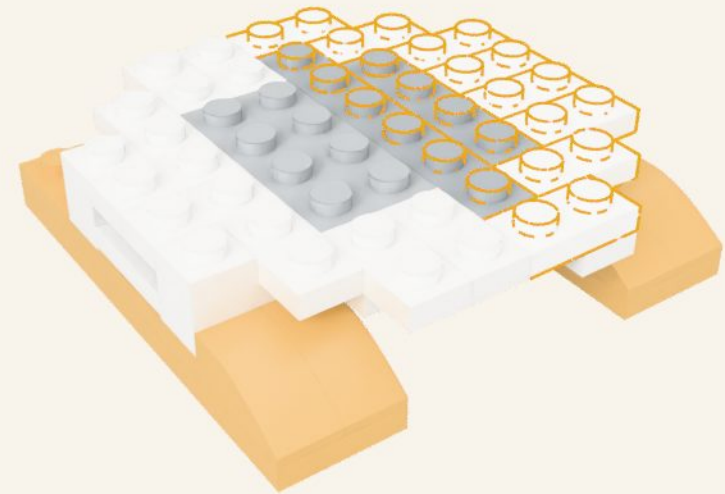
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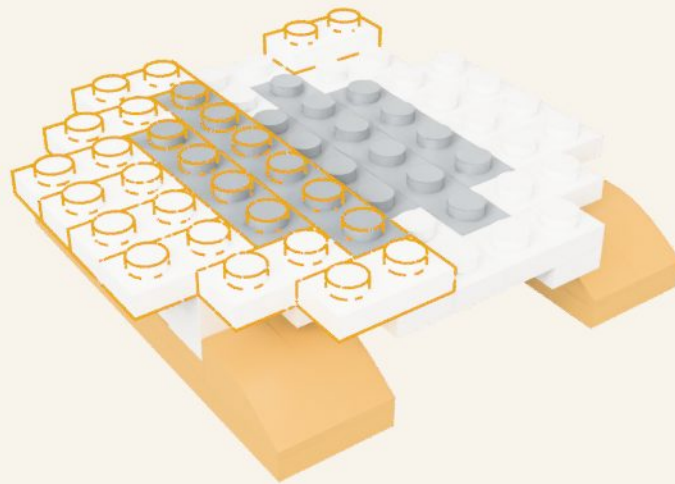
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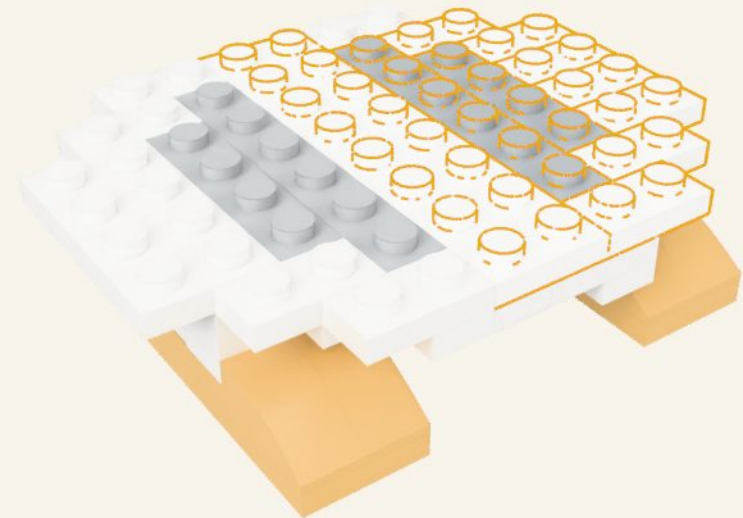
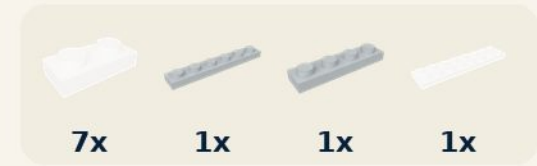
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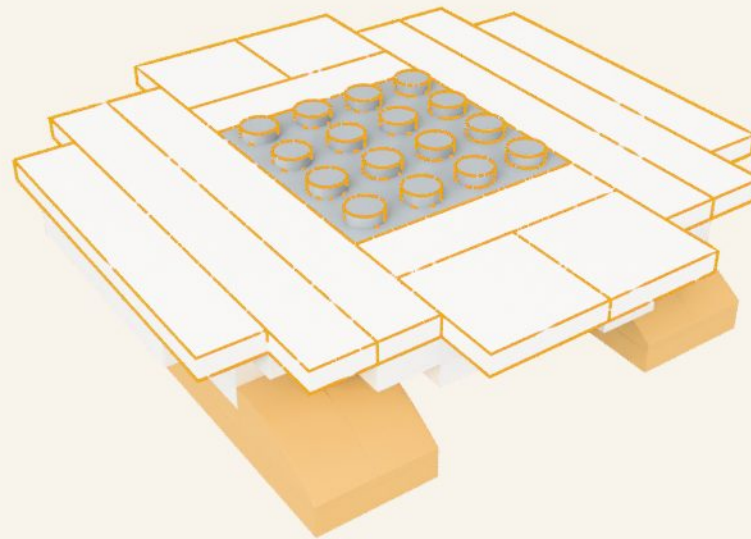
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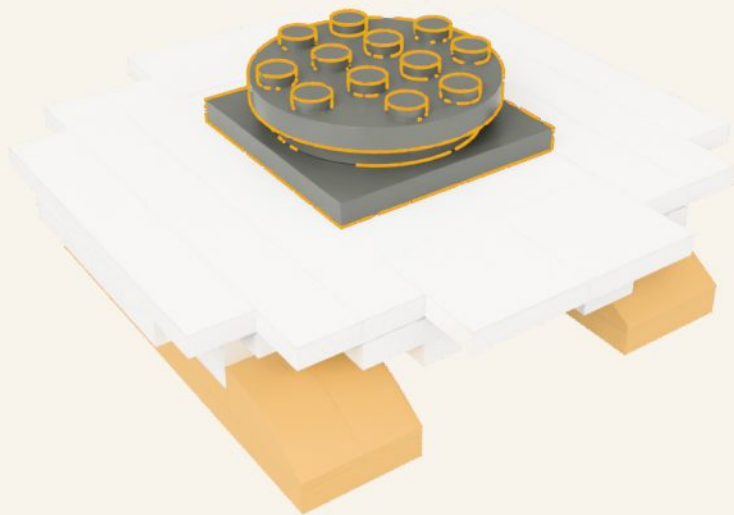


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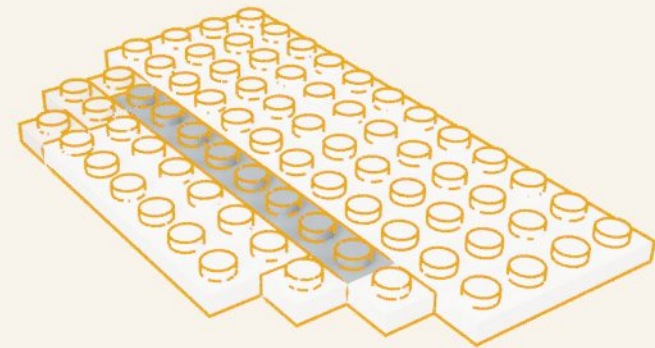


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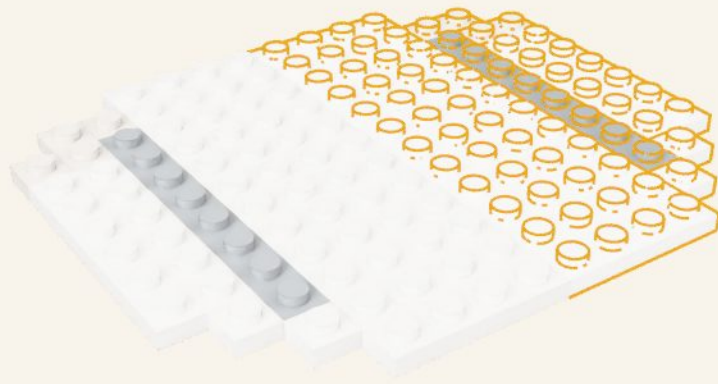
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Turntable: the upper body can turn on its feet

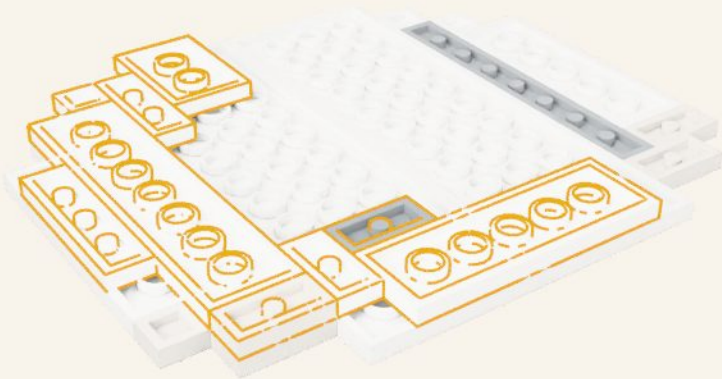
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Build the upper module separately

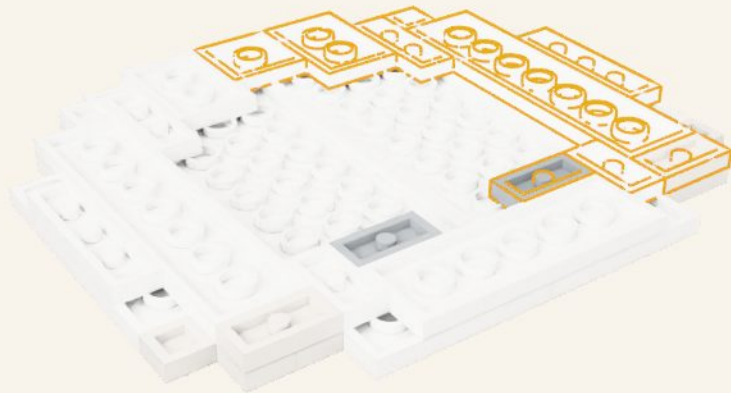
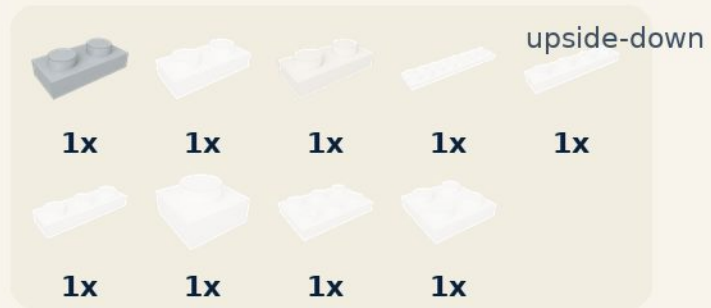
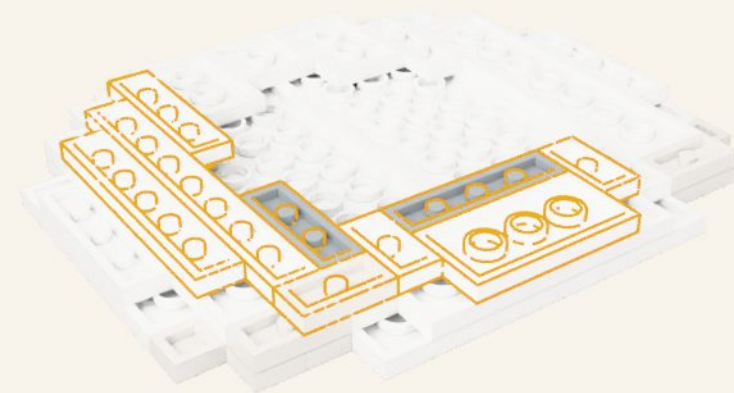
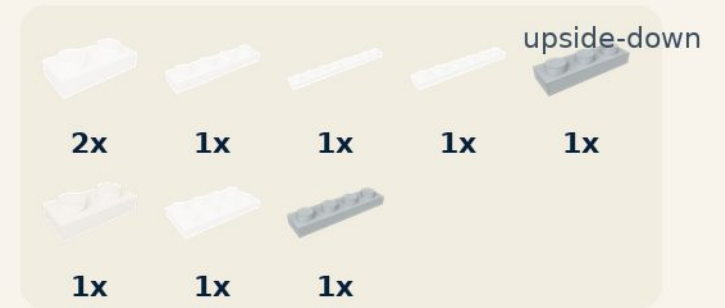
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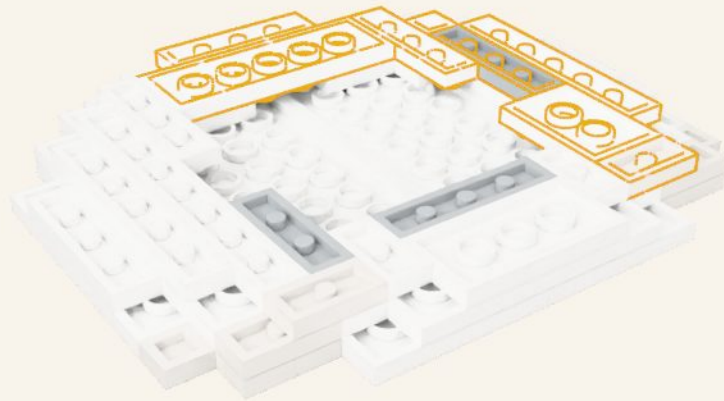
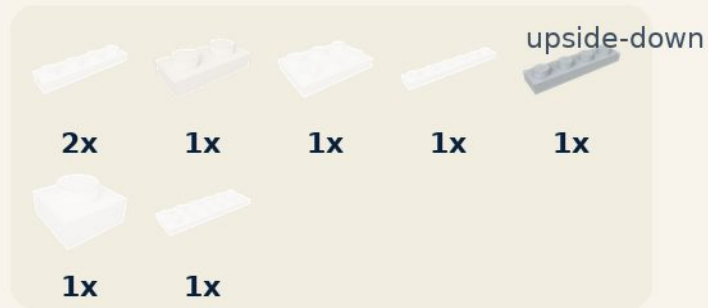
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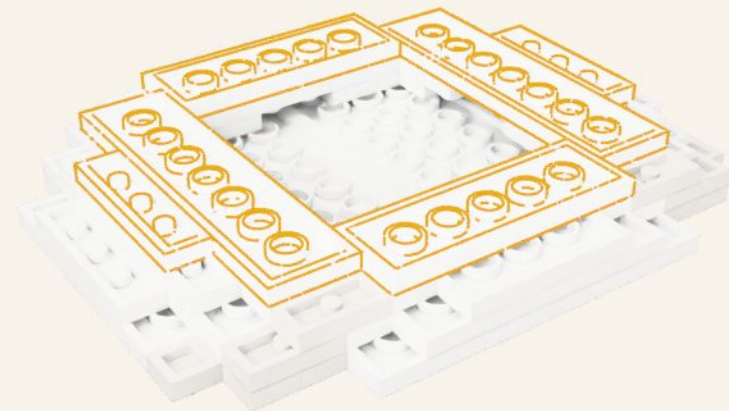
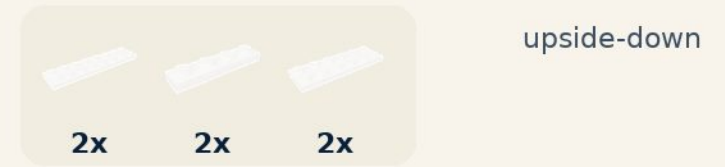
Turn the module over

14**15**

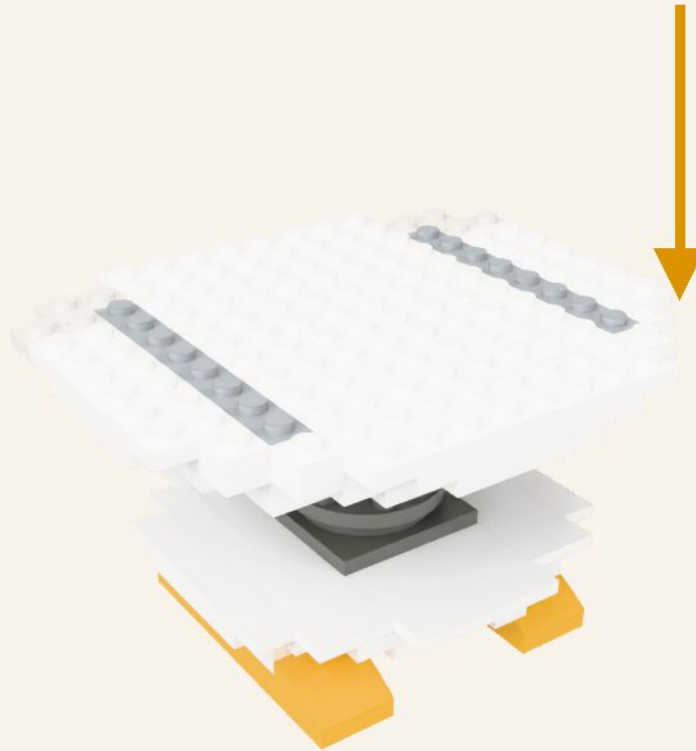
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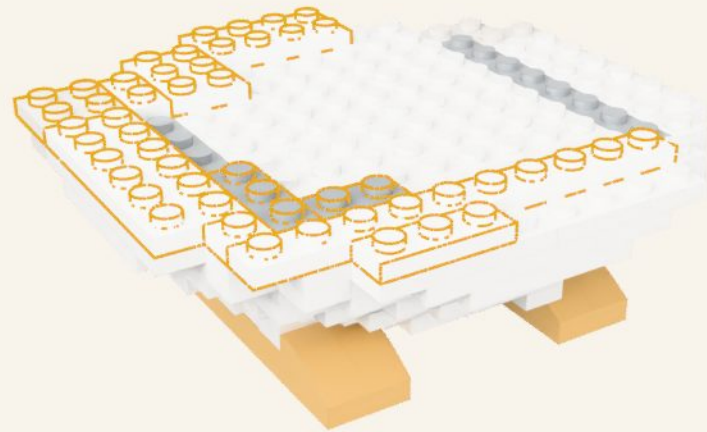


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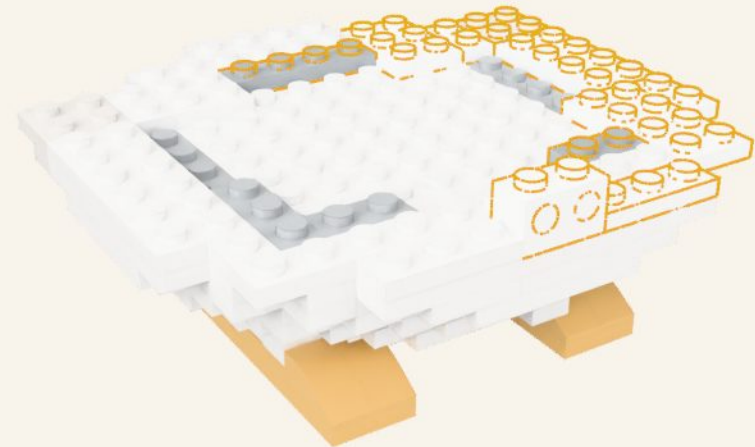


Turn the module back and press it onto the turntable

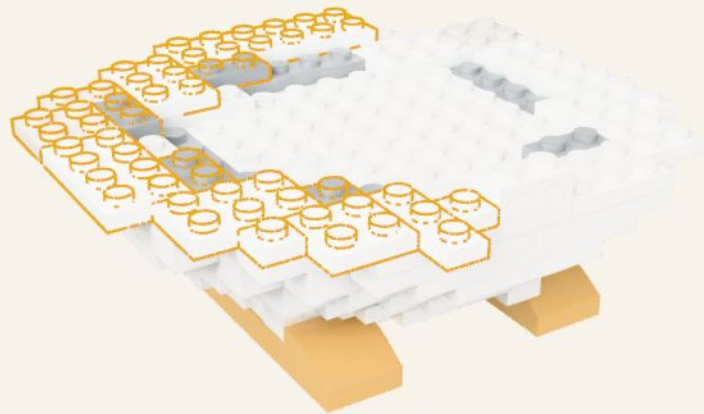
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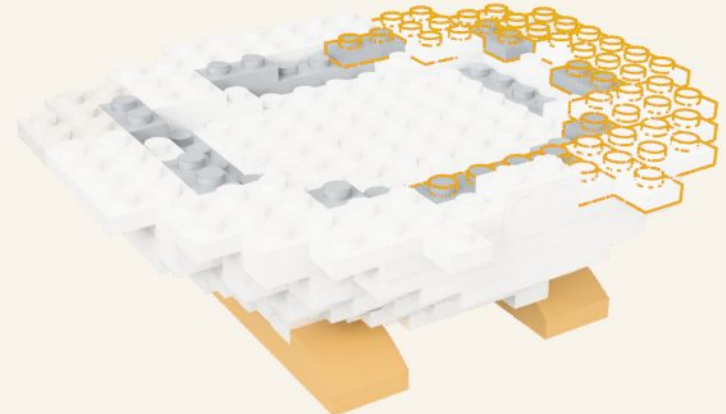
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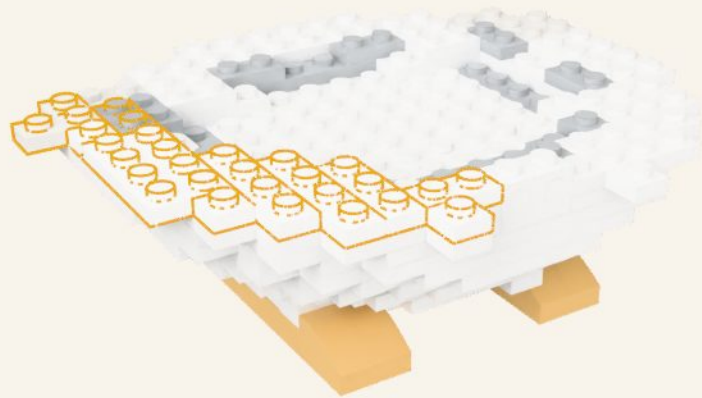
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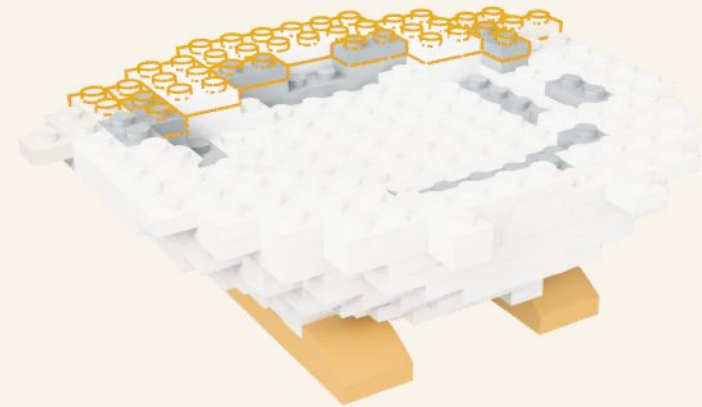
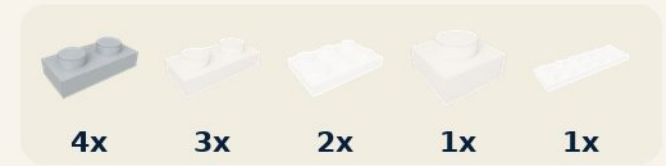
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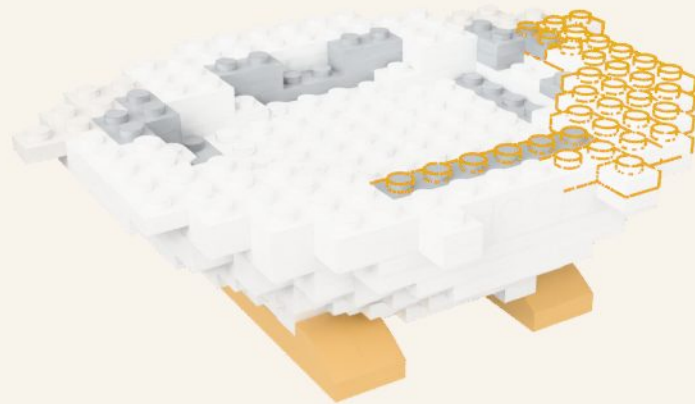
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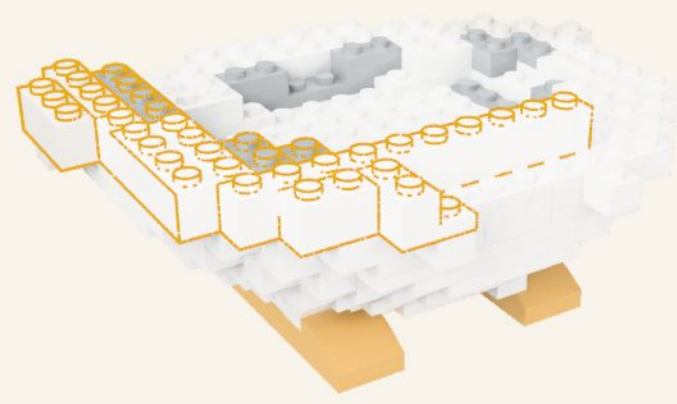
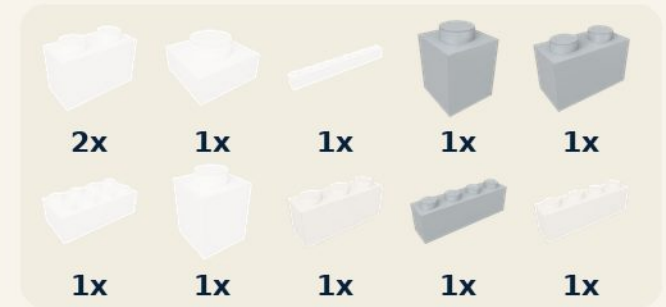
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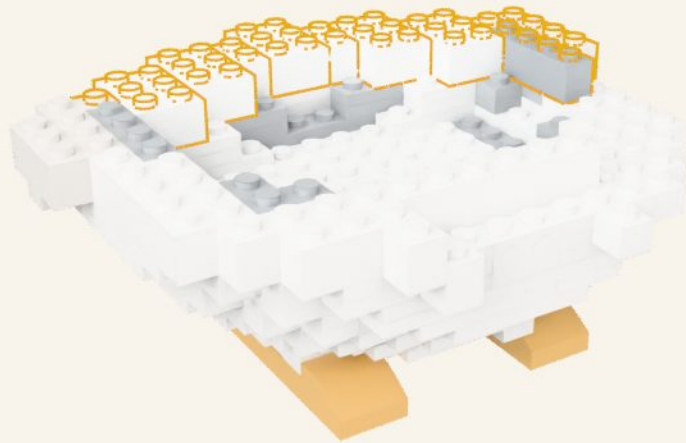
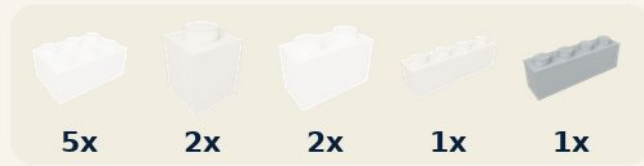
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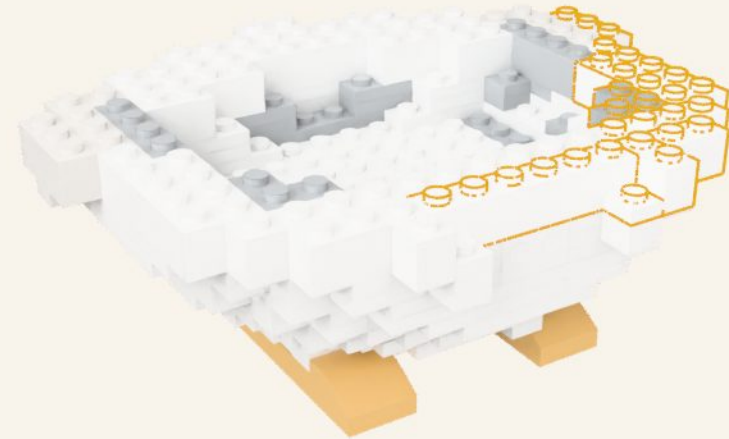
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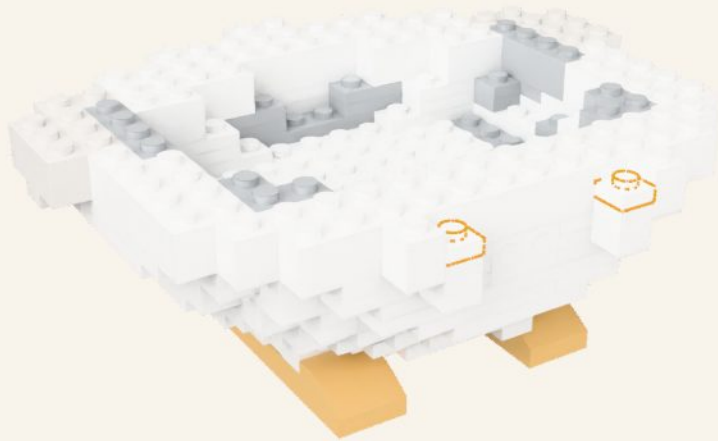
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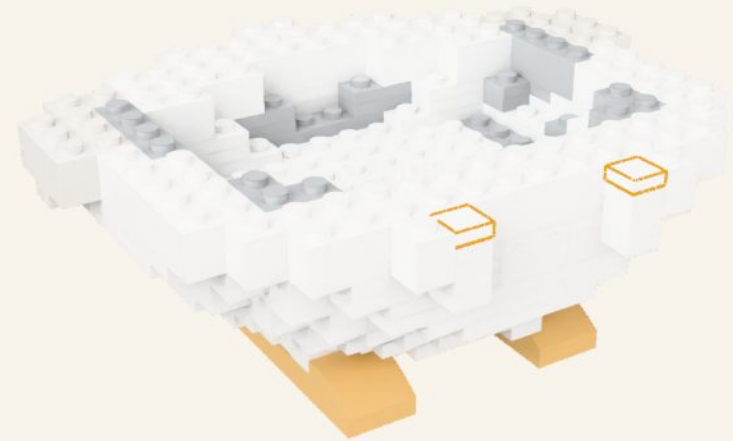
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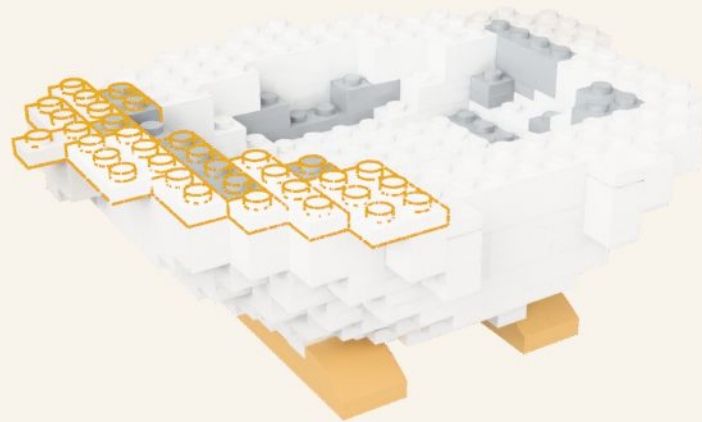
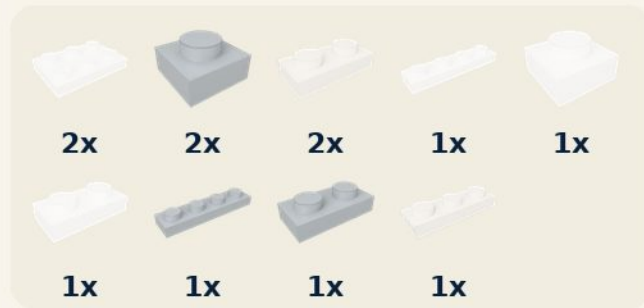
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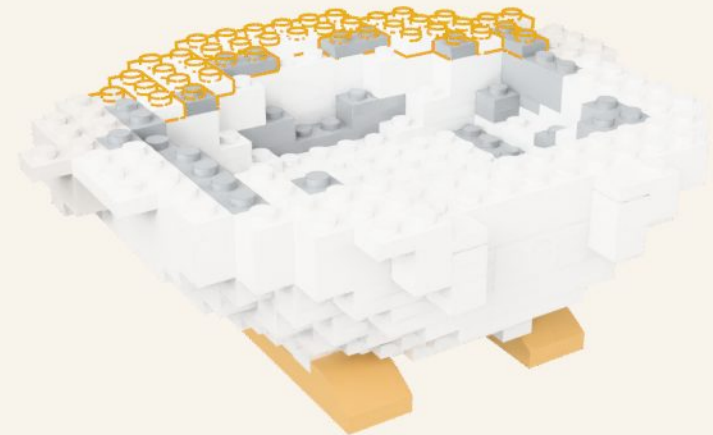
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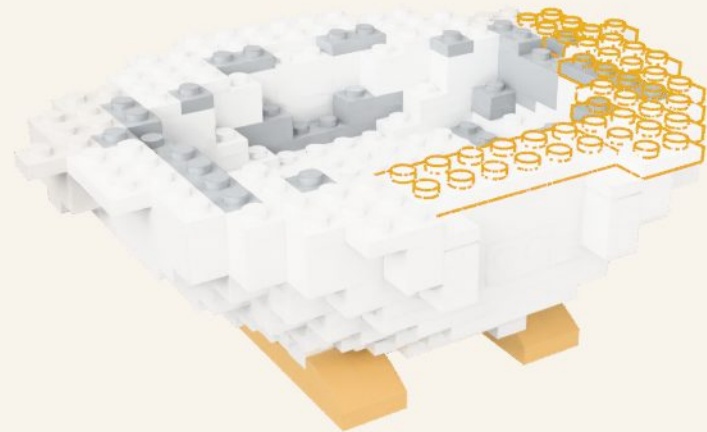
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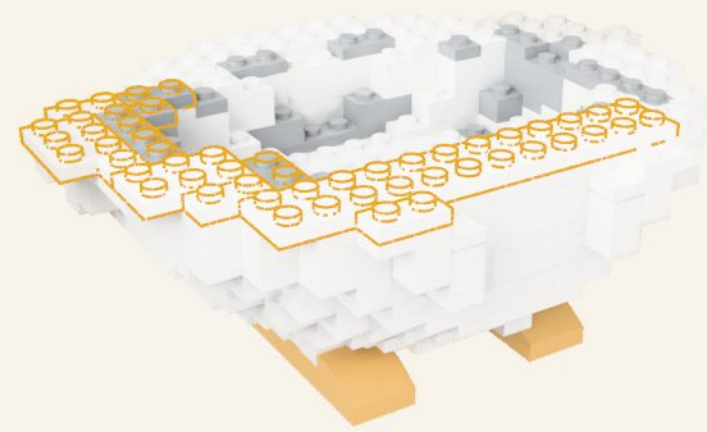
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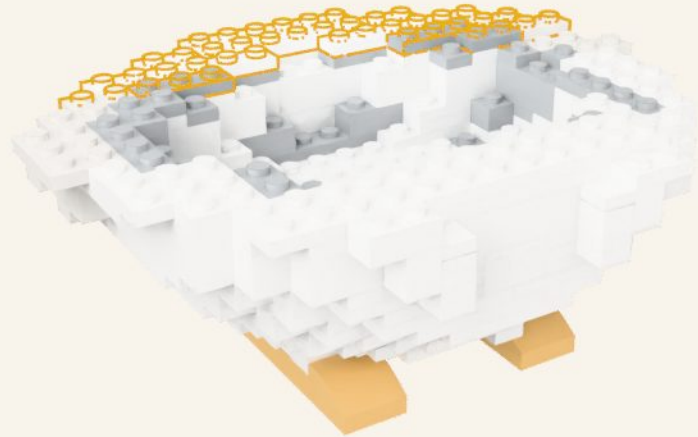
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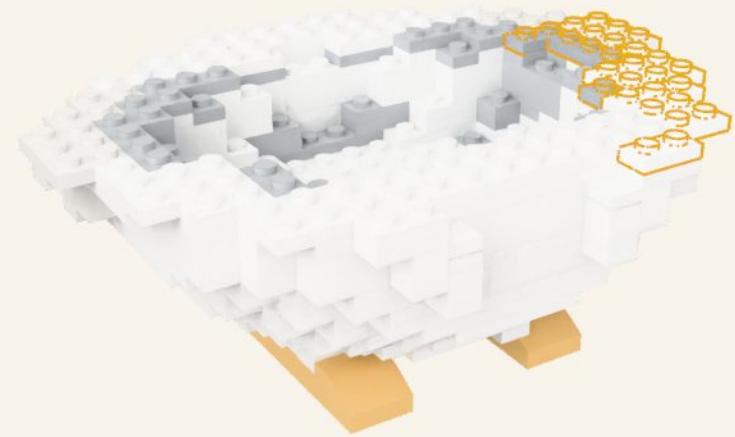
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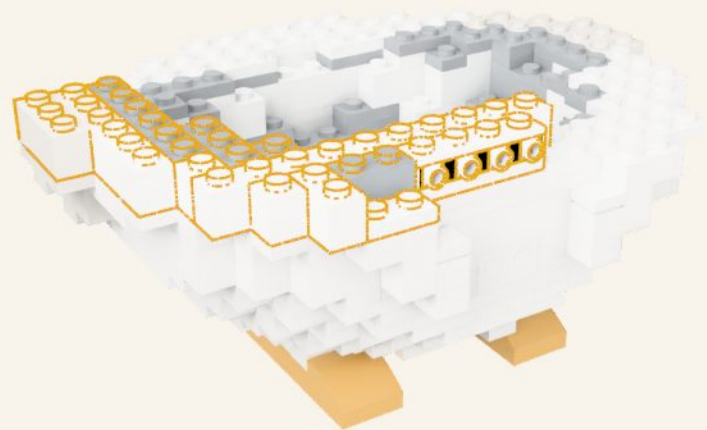
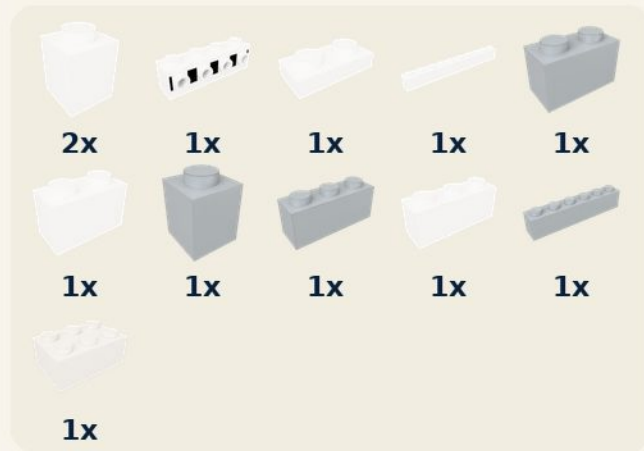
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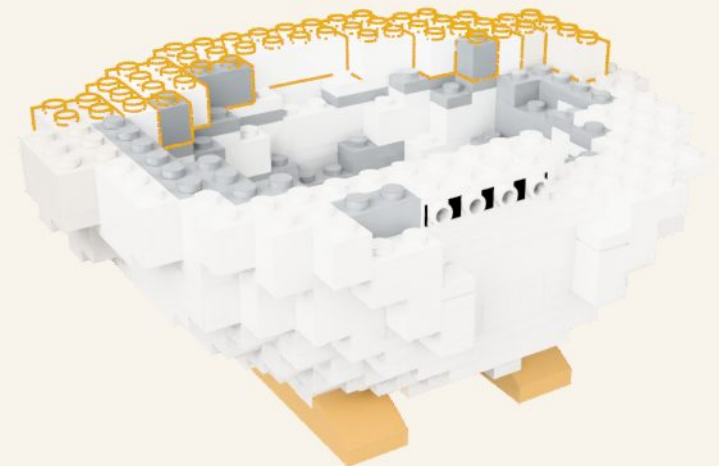
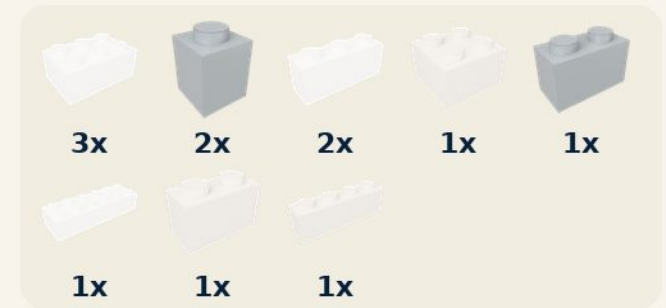
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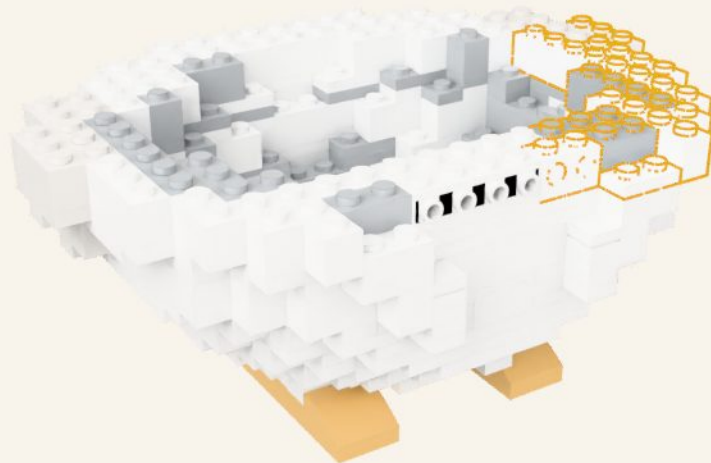
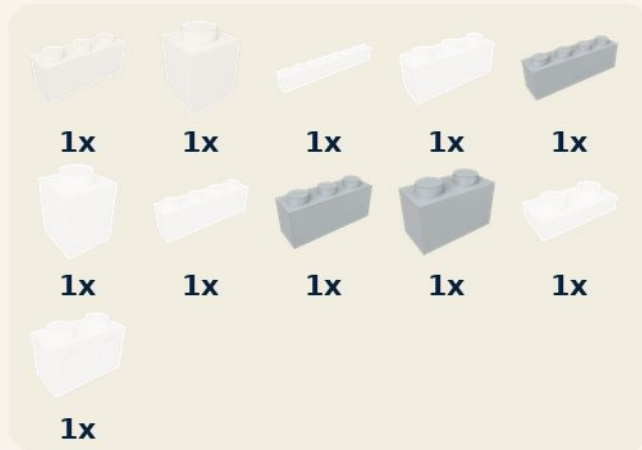
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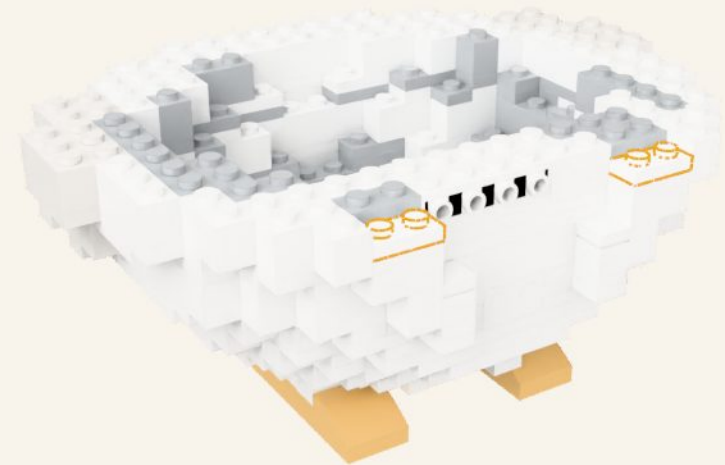
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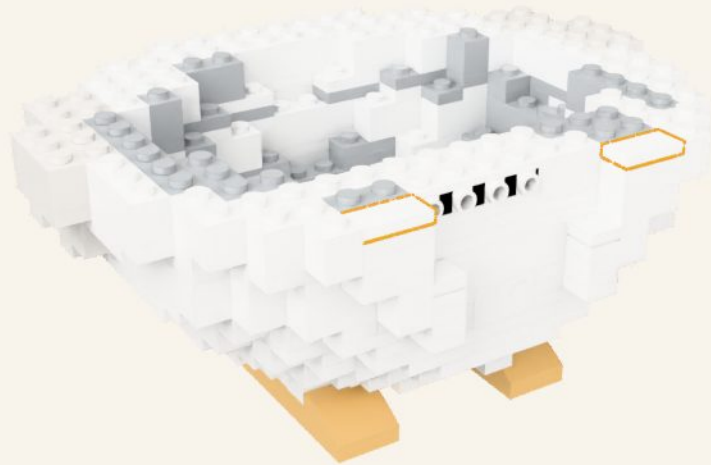
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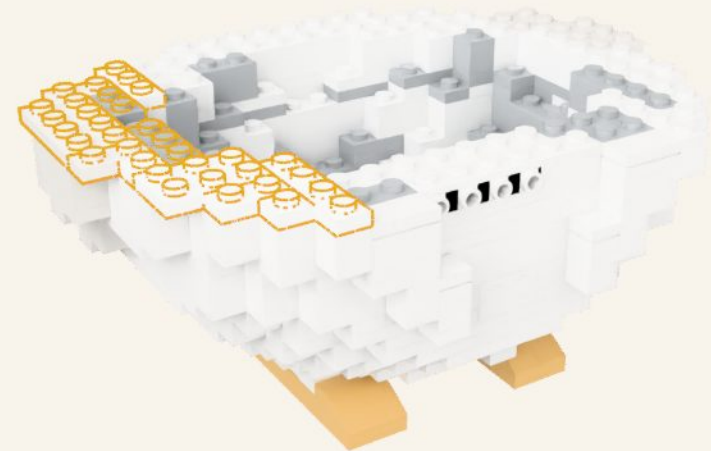
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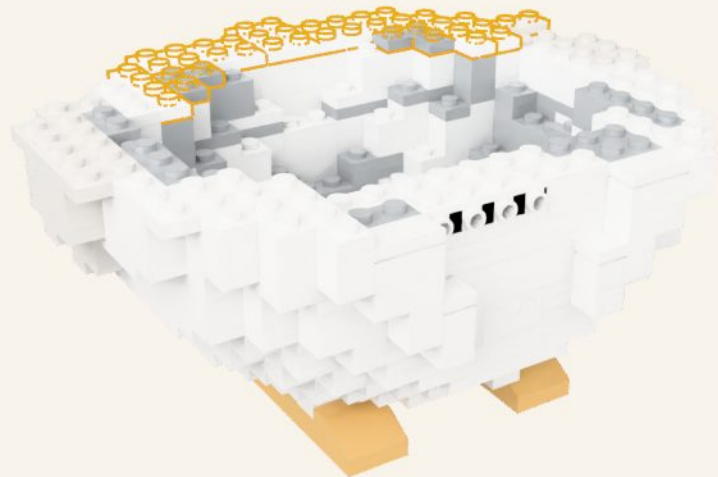
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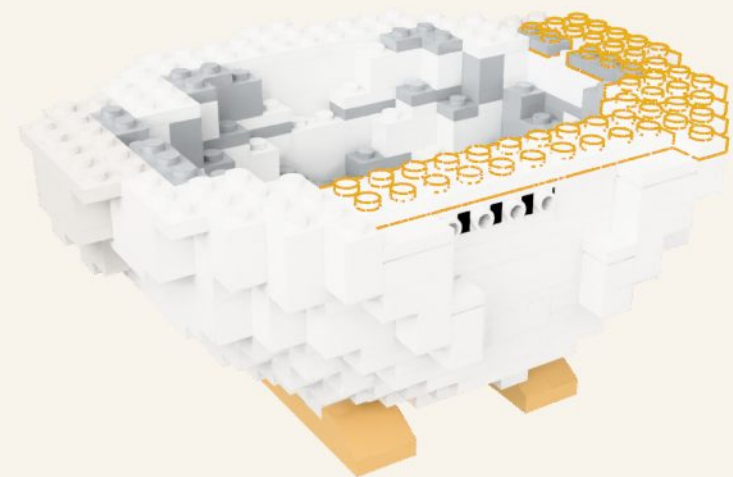
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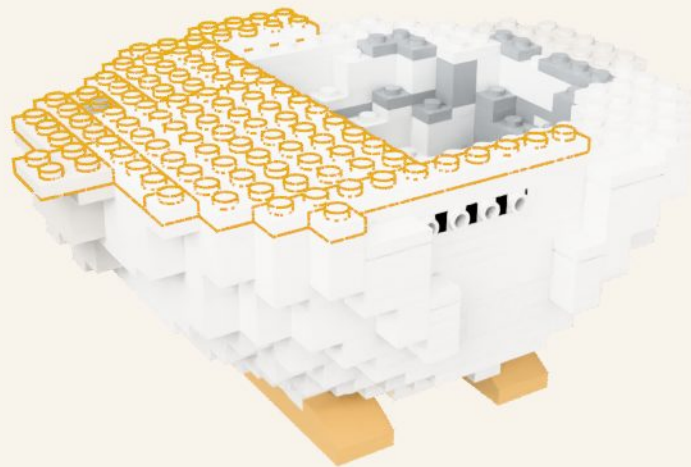
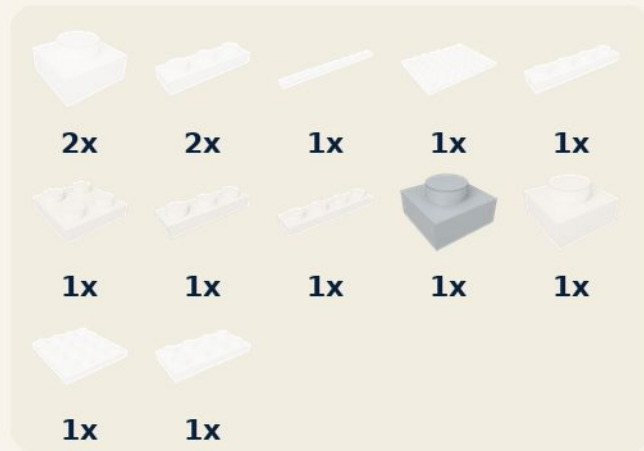
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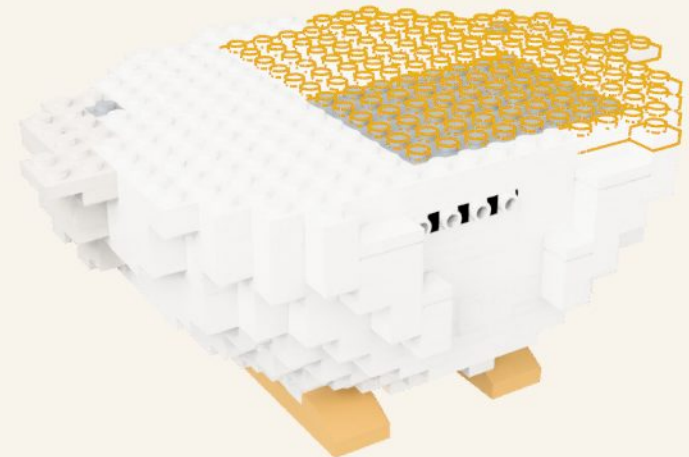
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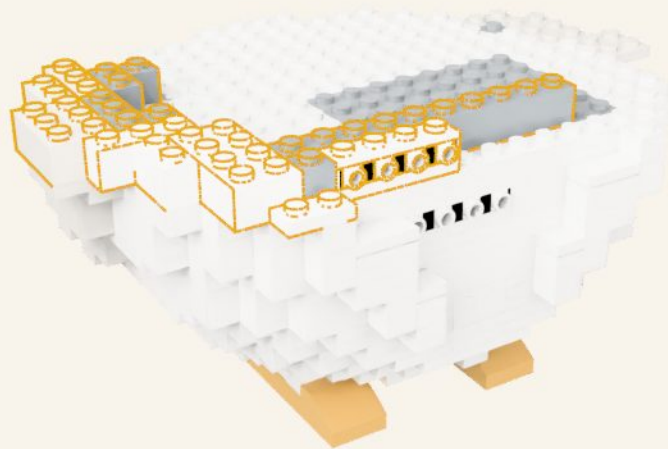
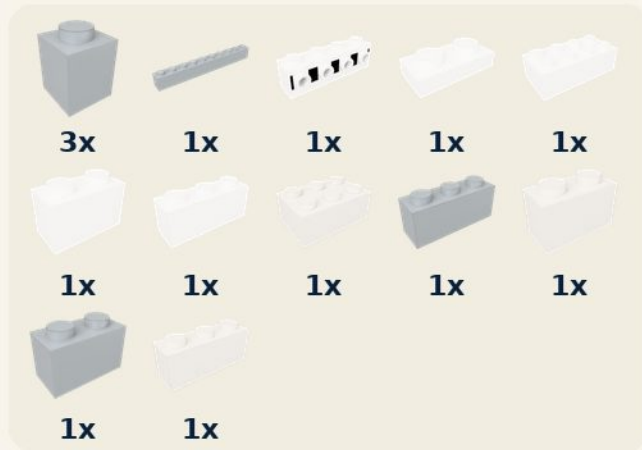
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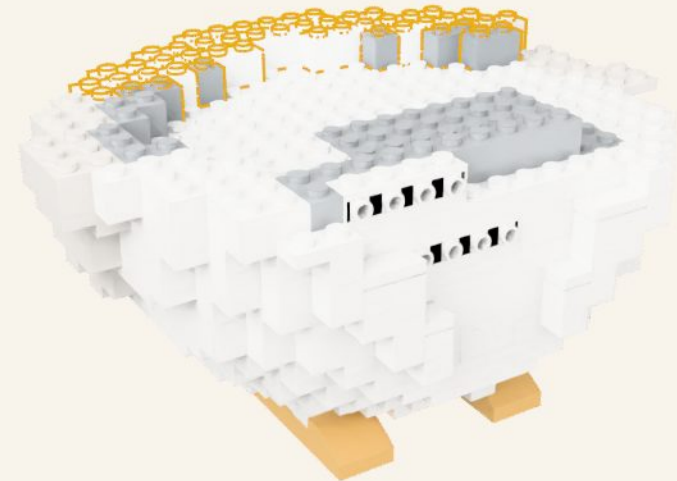
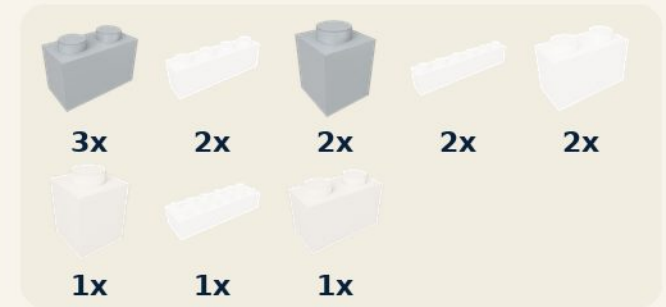
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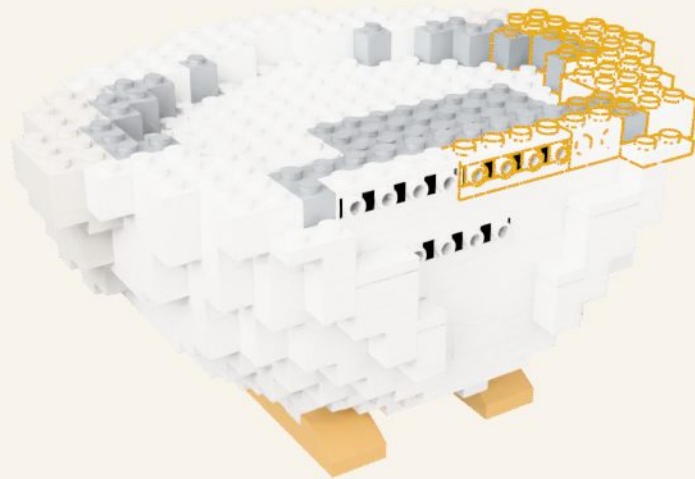
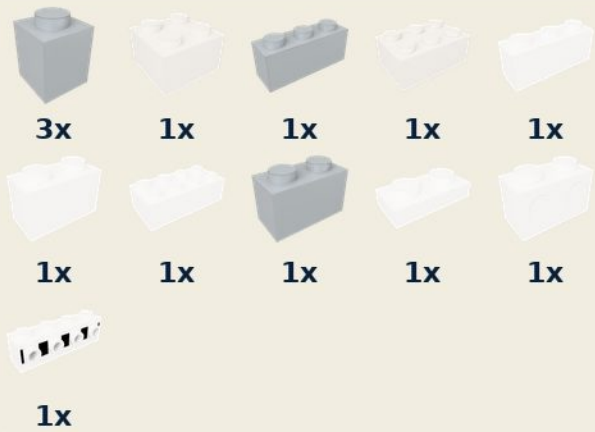
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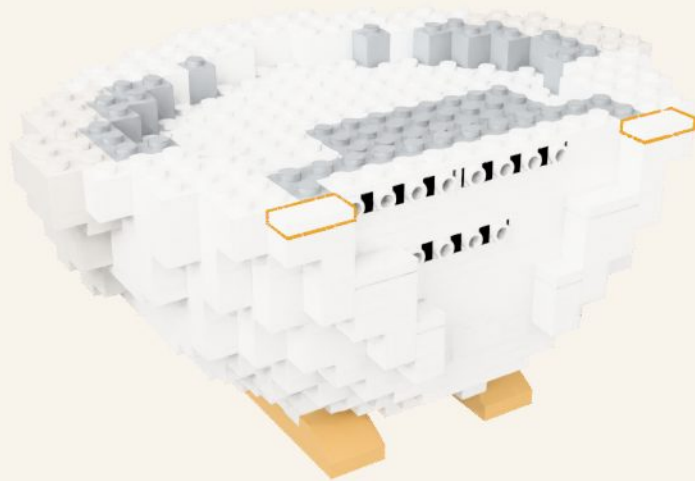
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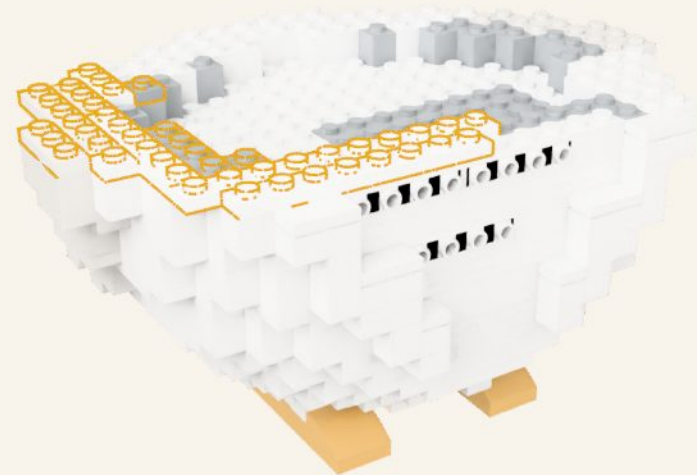
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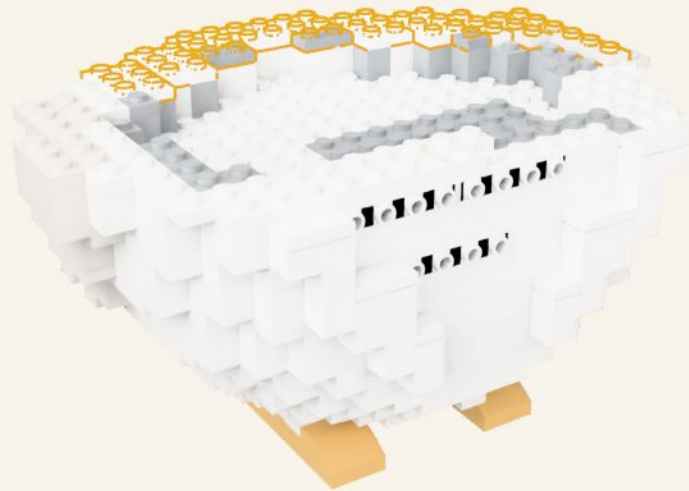
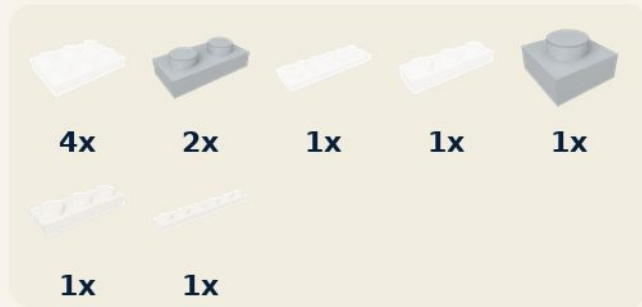
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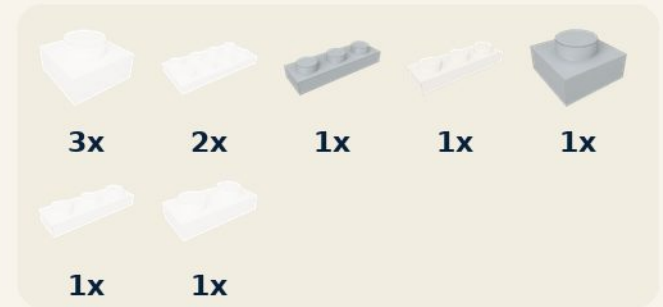
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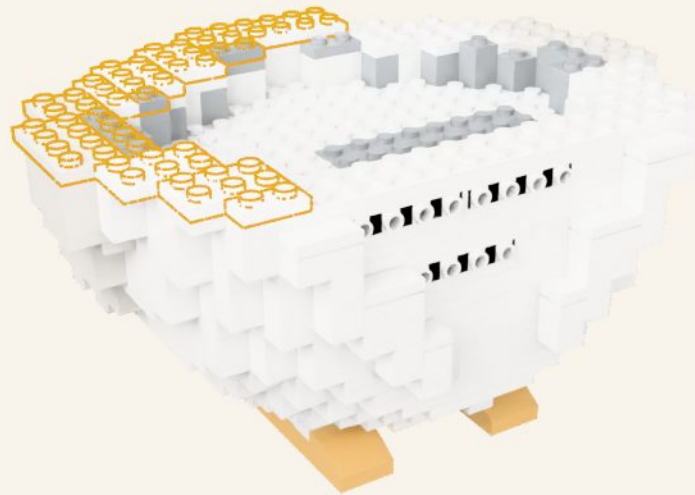
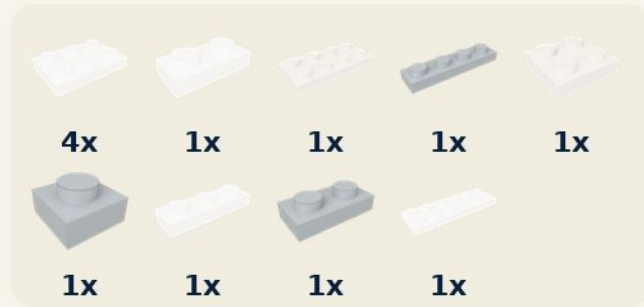
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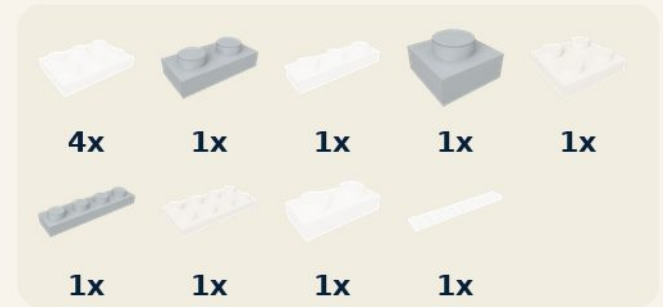
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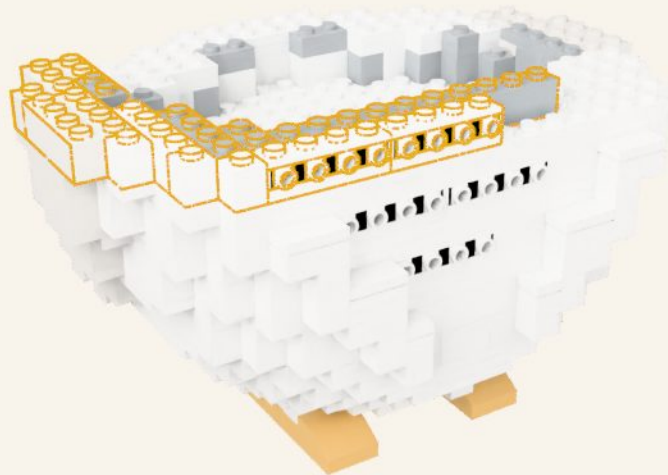
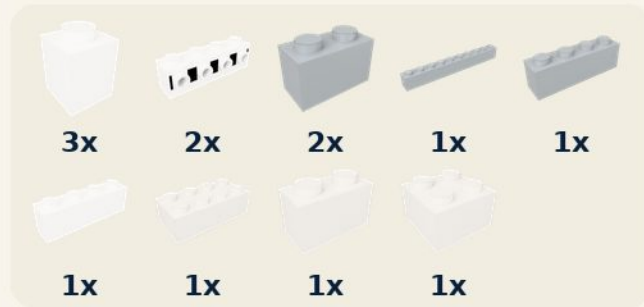
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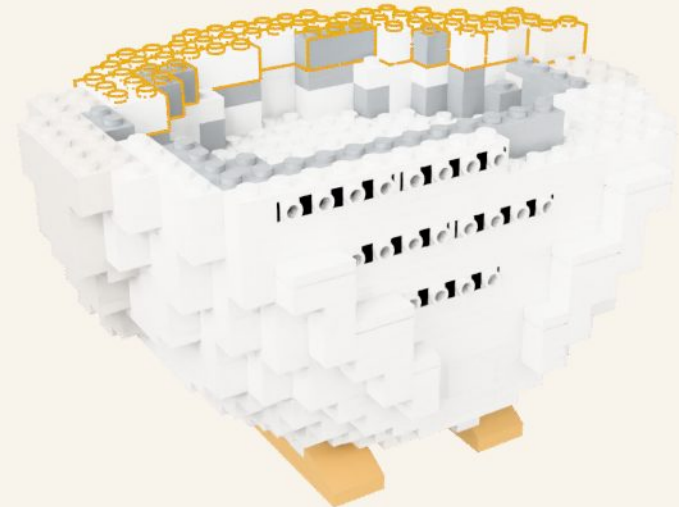
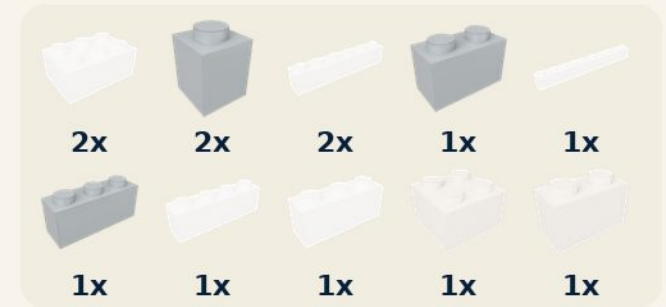
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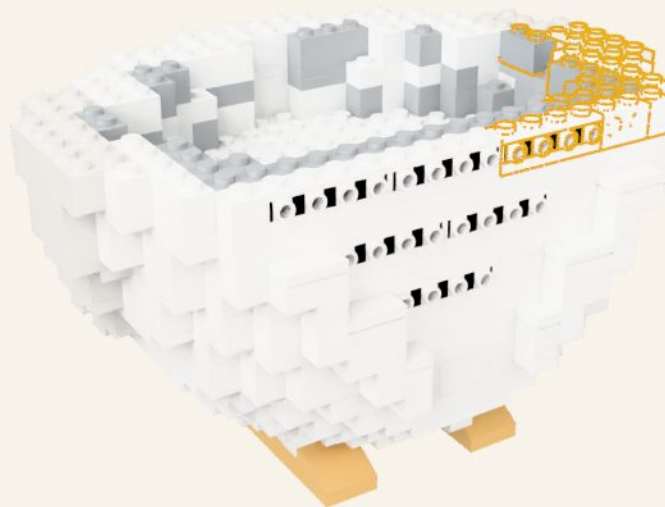
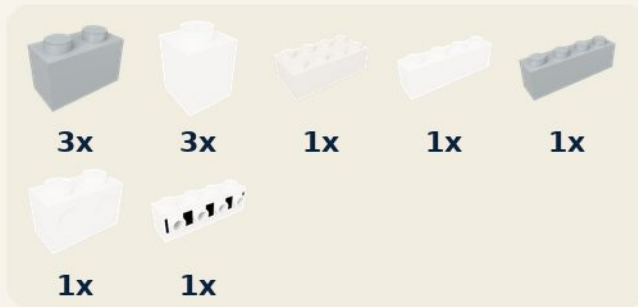
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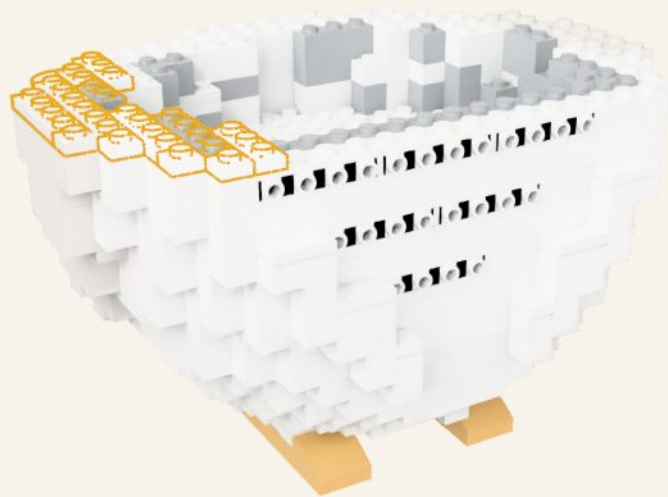
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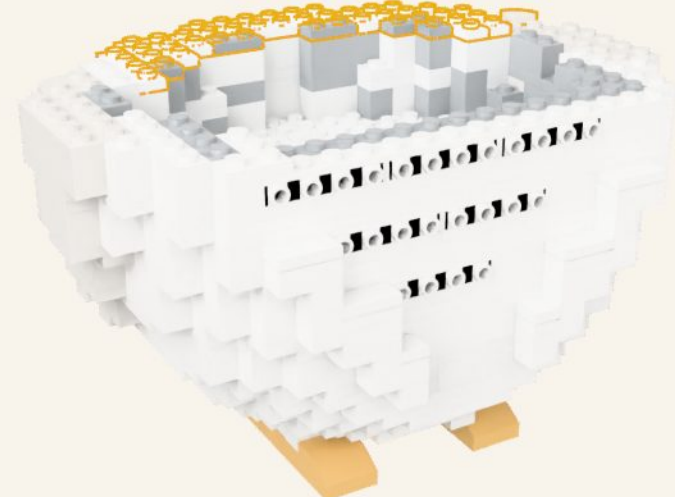
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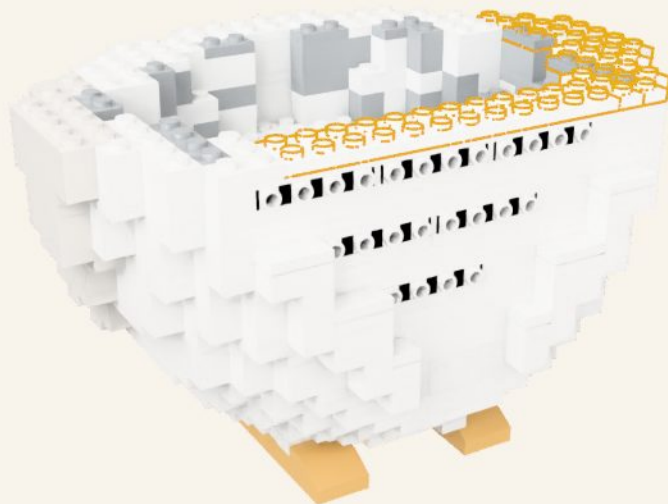
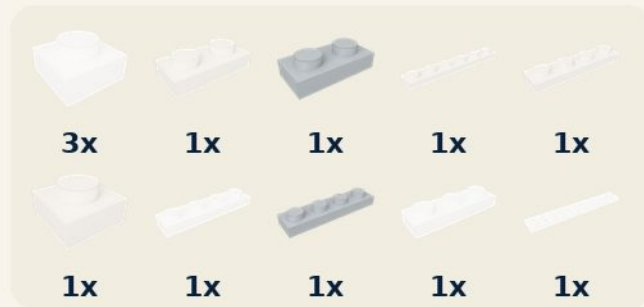
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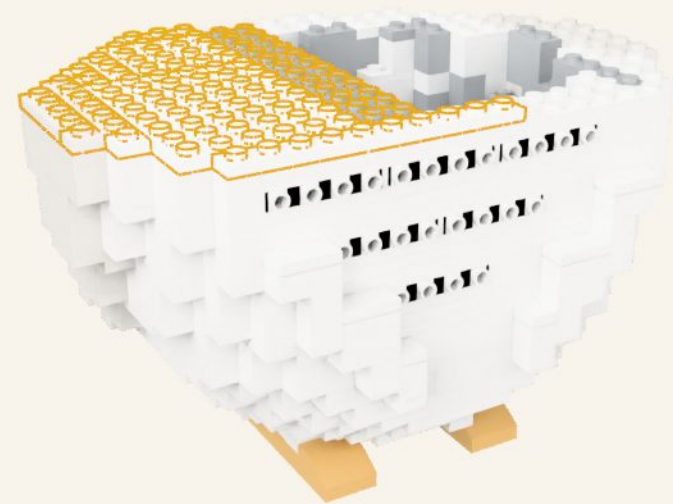
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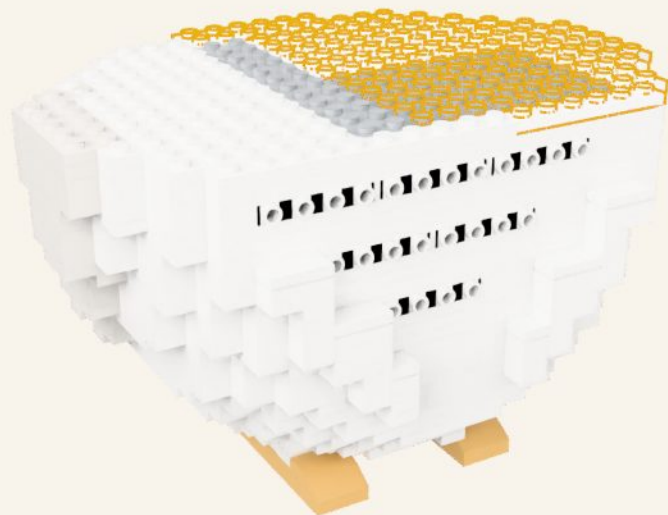
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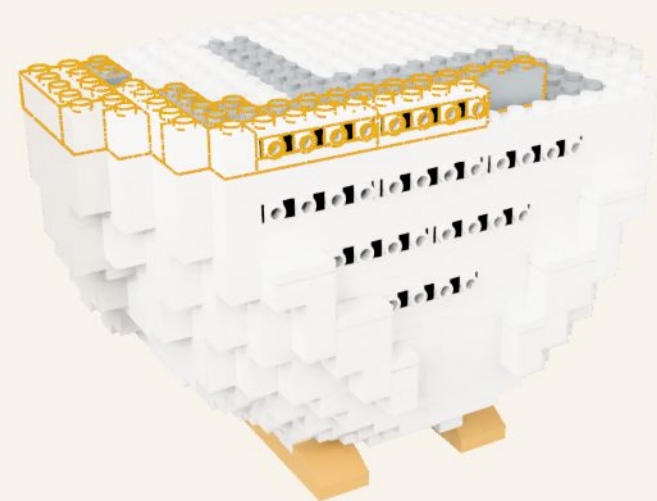
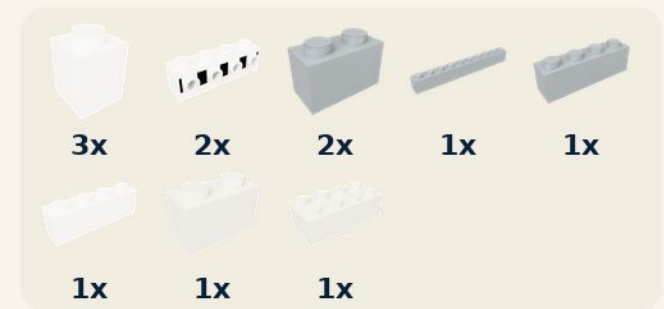
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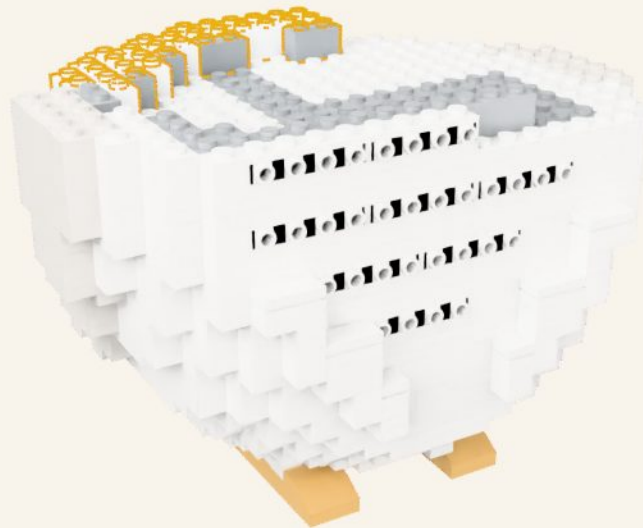
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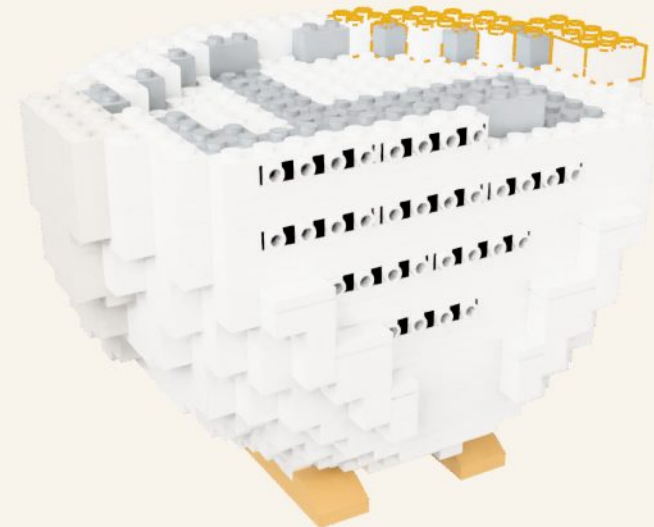
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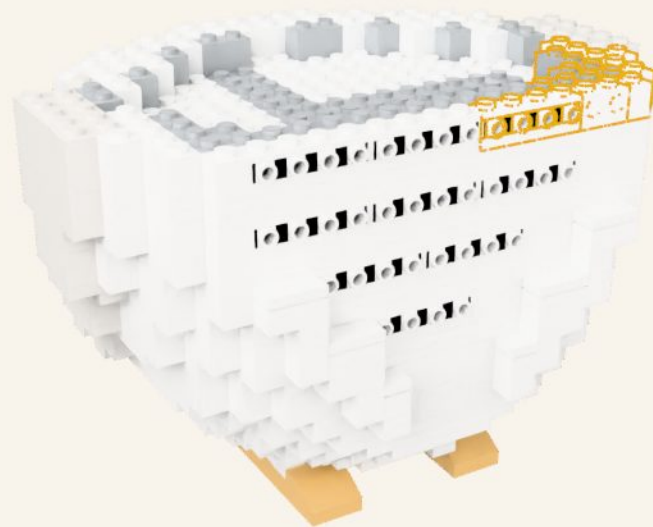
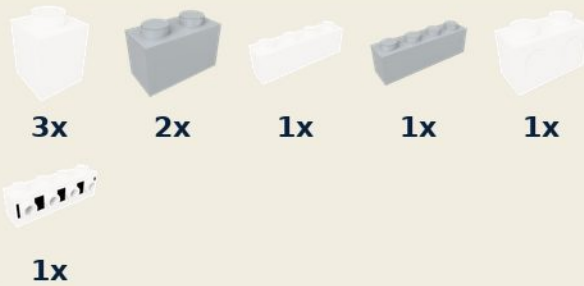
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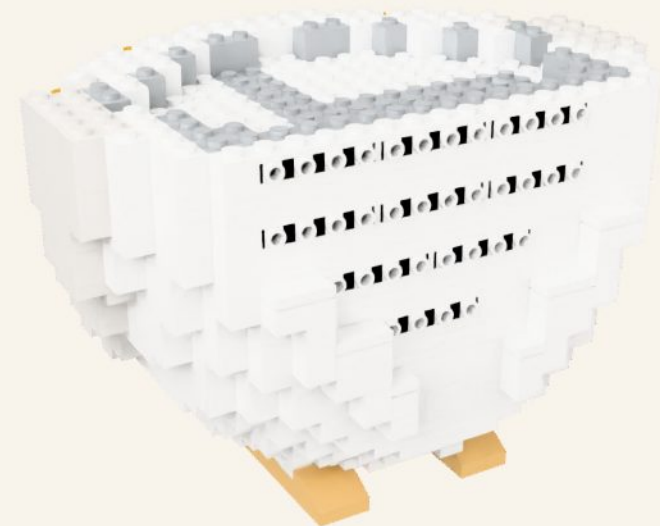
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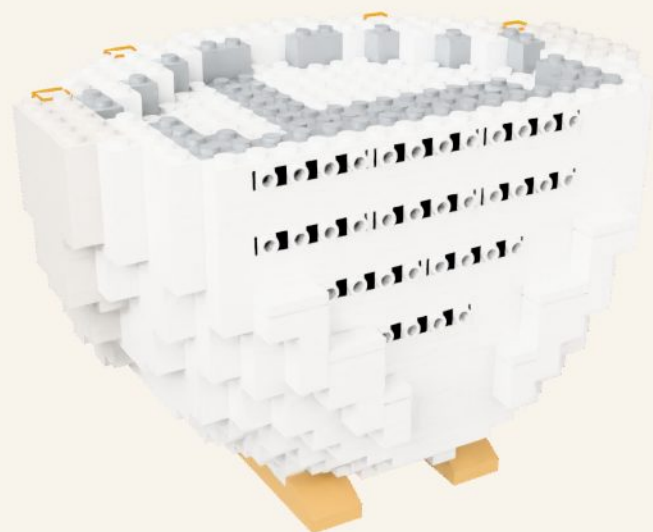
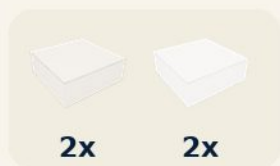
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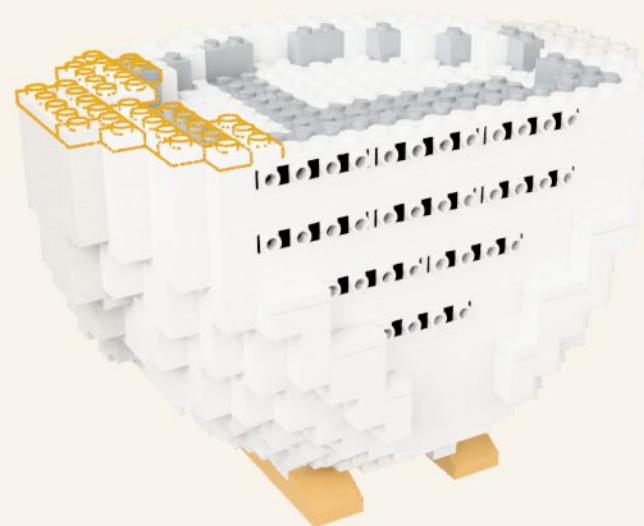
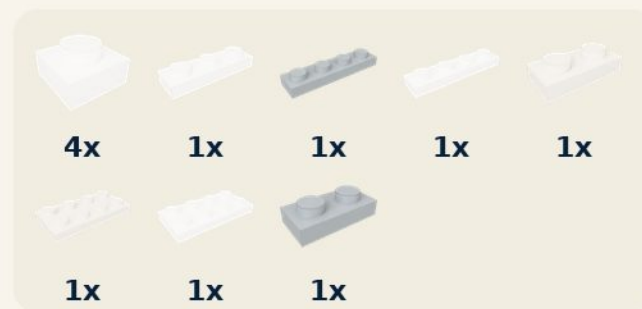
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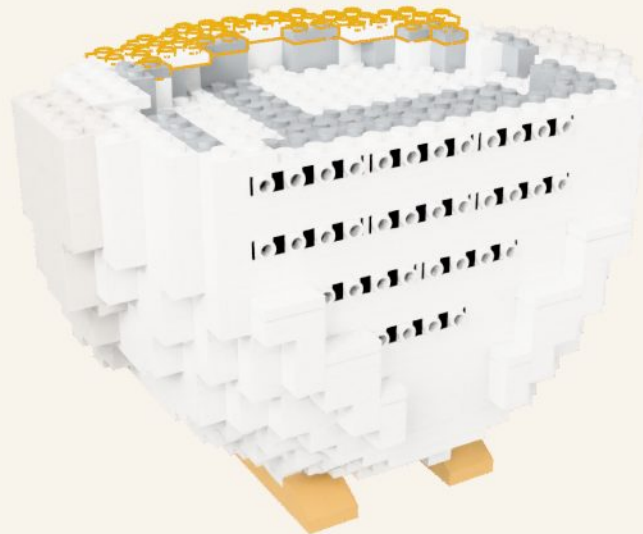
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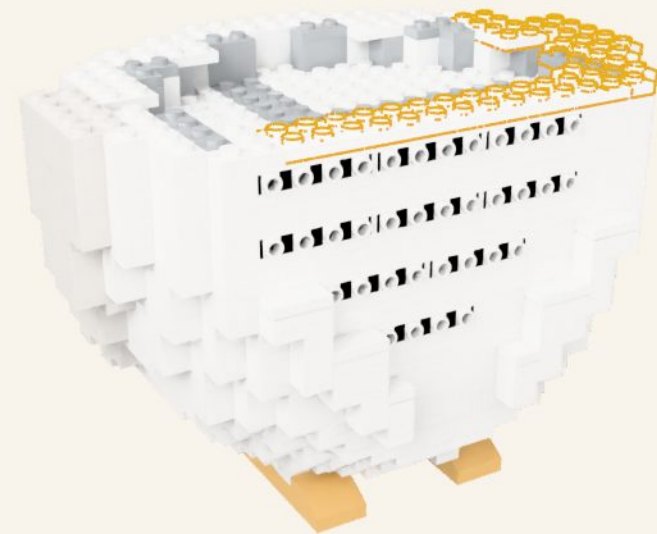
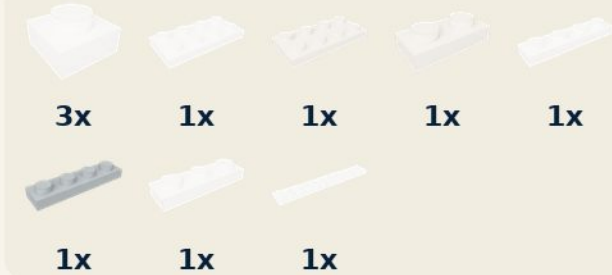
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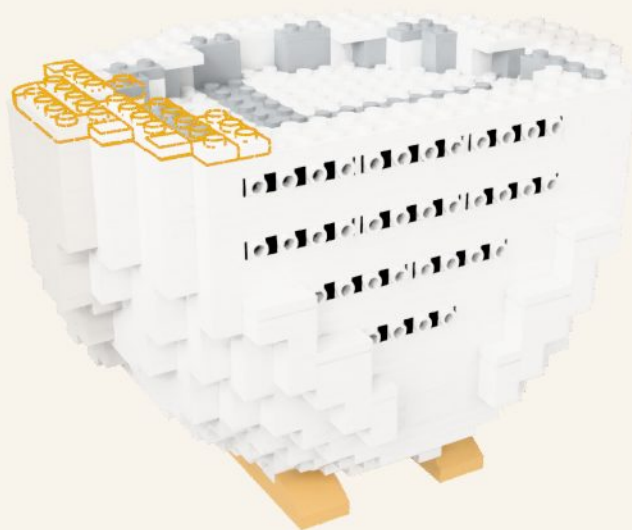
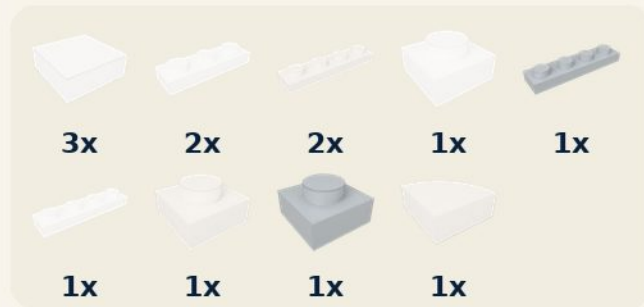
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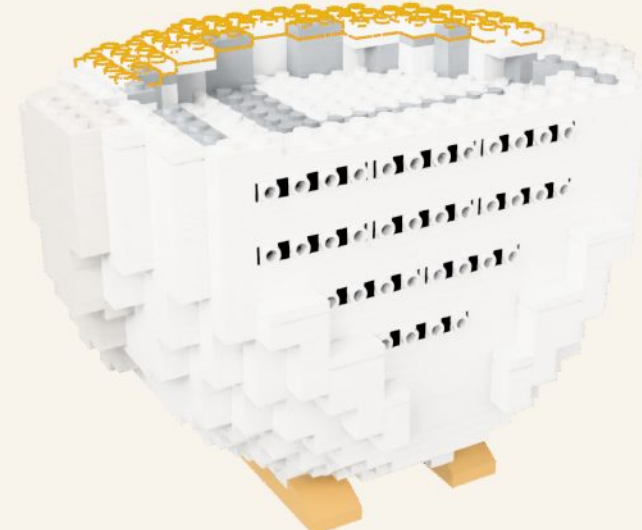
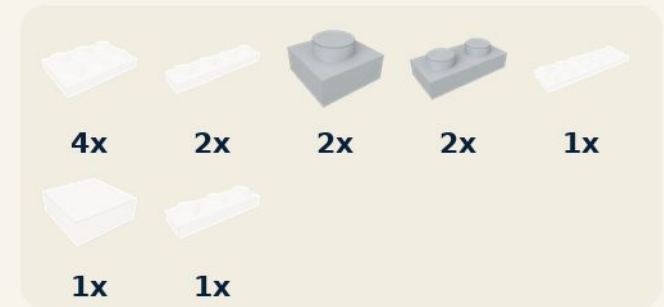
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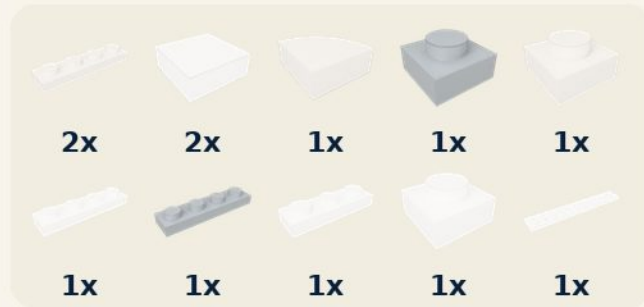
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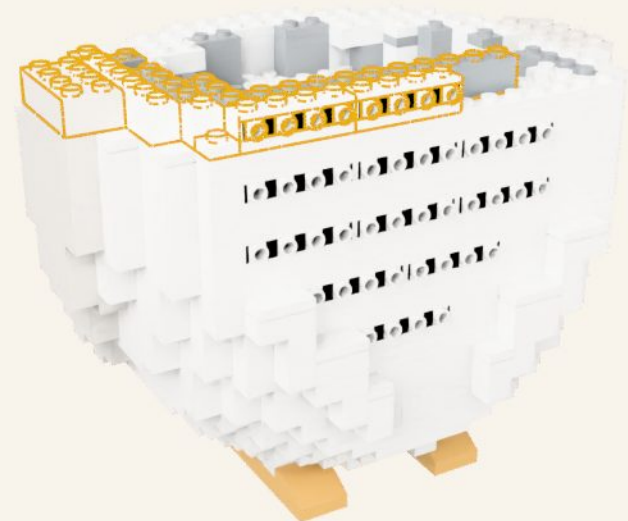
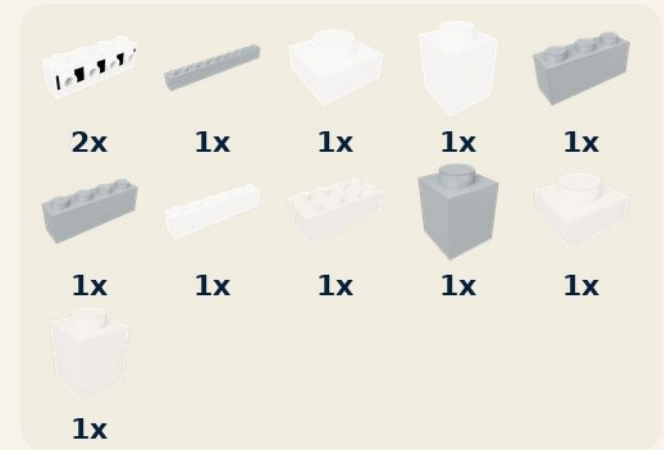
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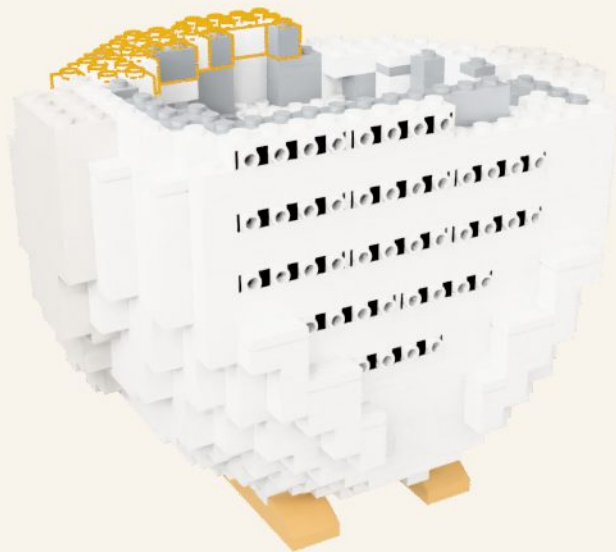
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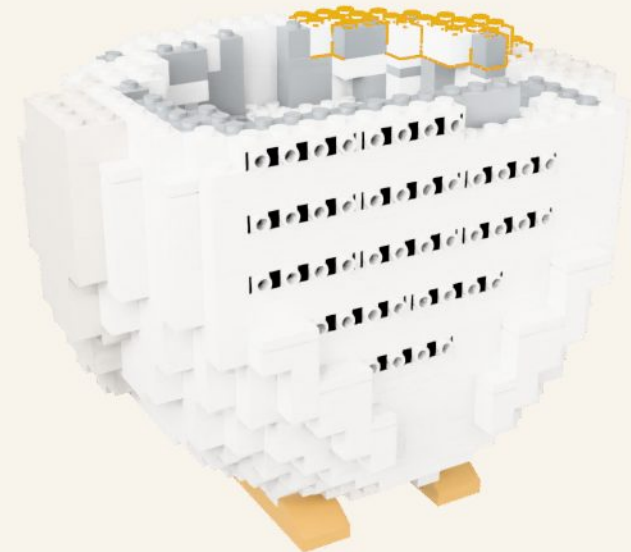
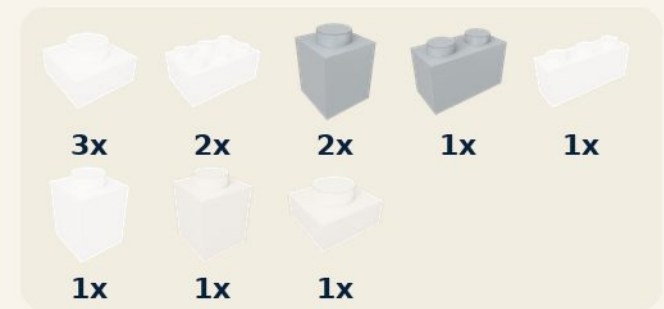
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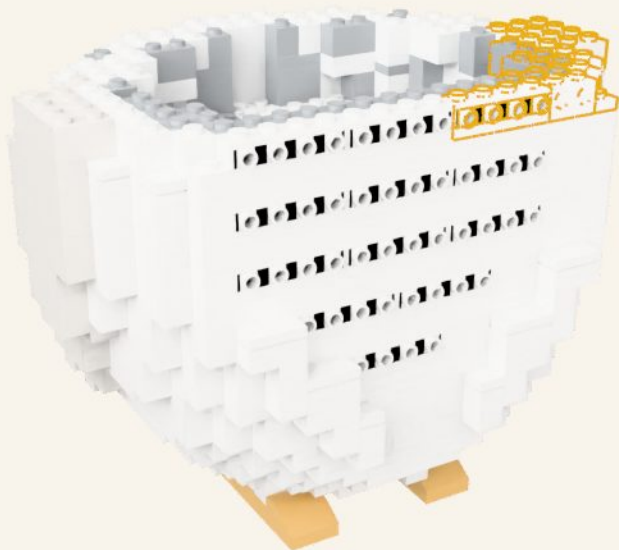
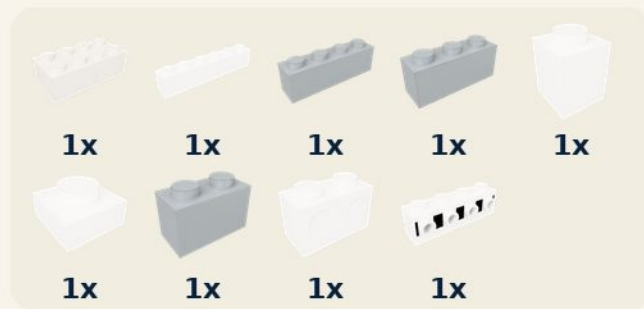
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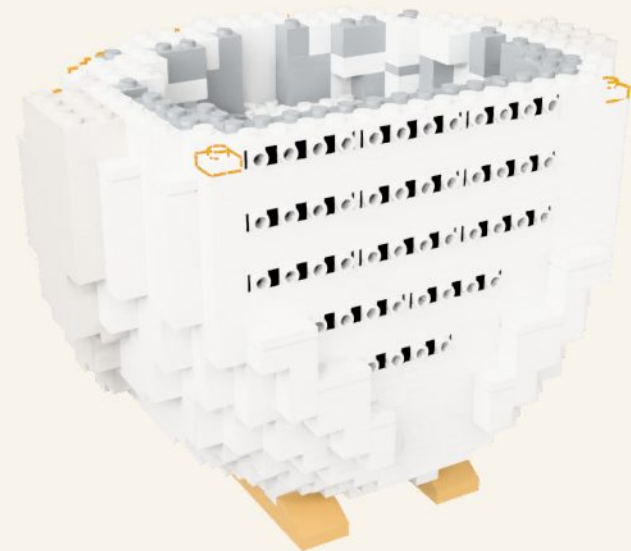
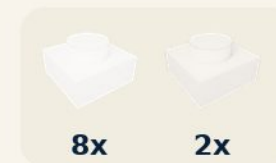
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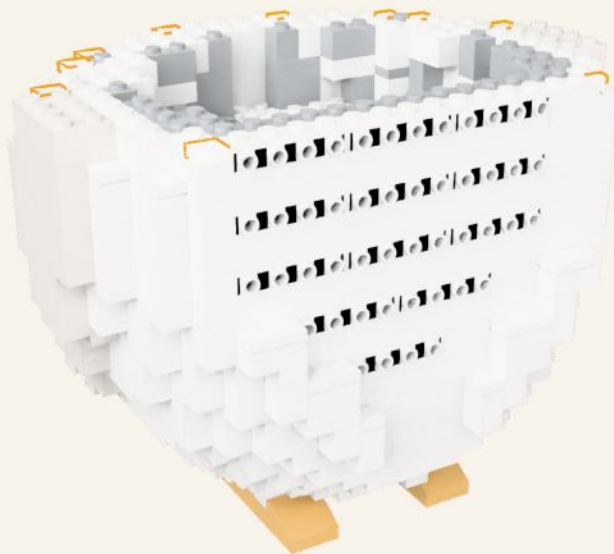
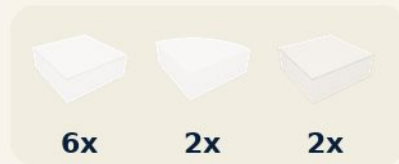
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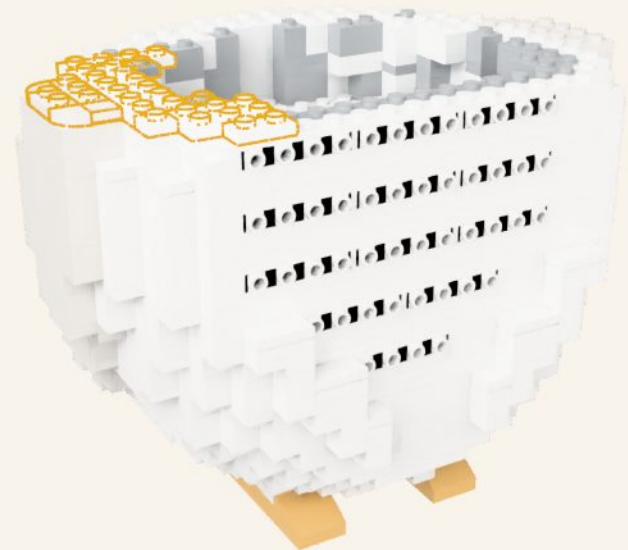
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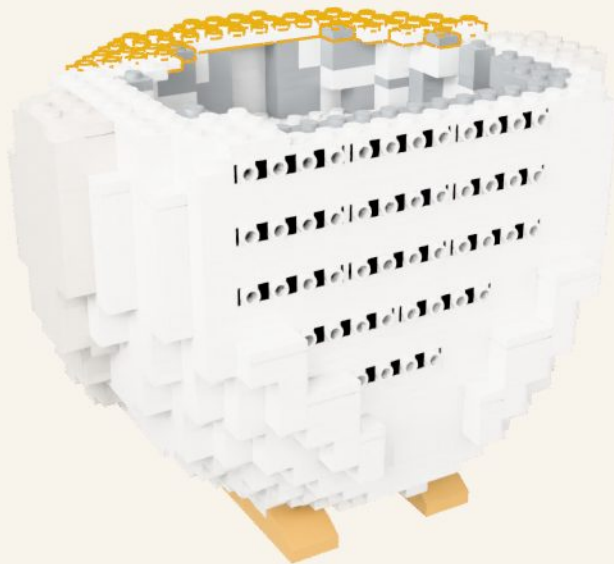
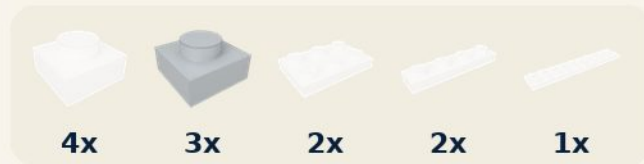
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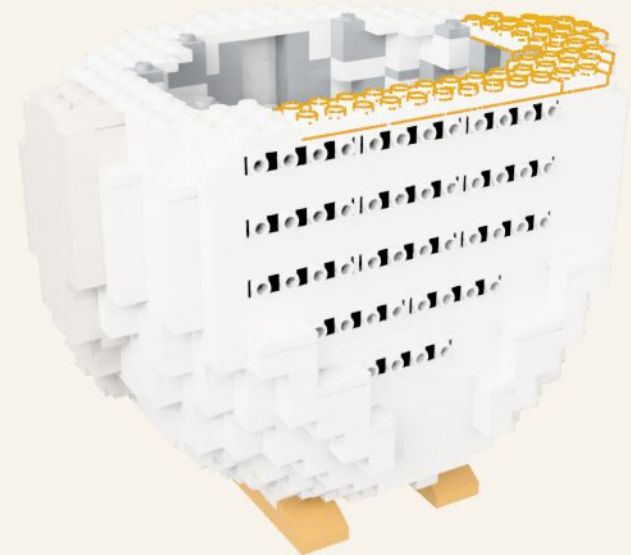
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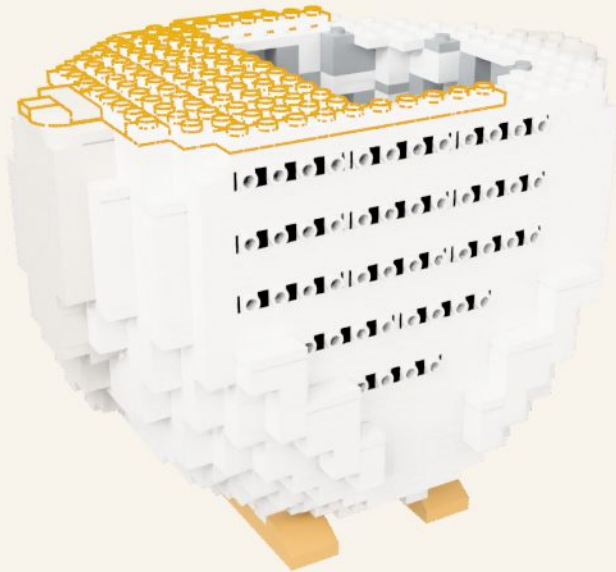
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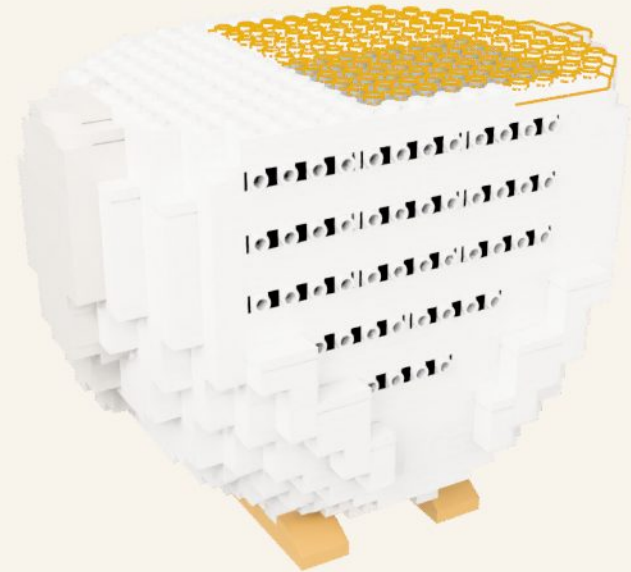
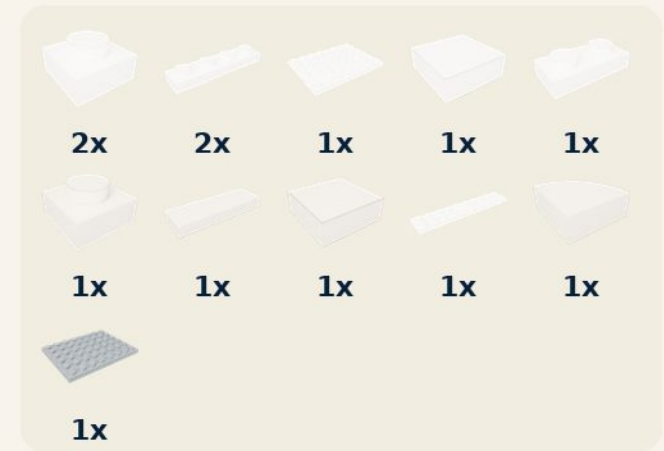
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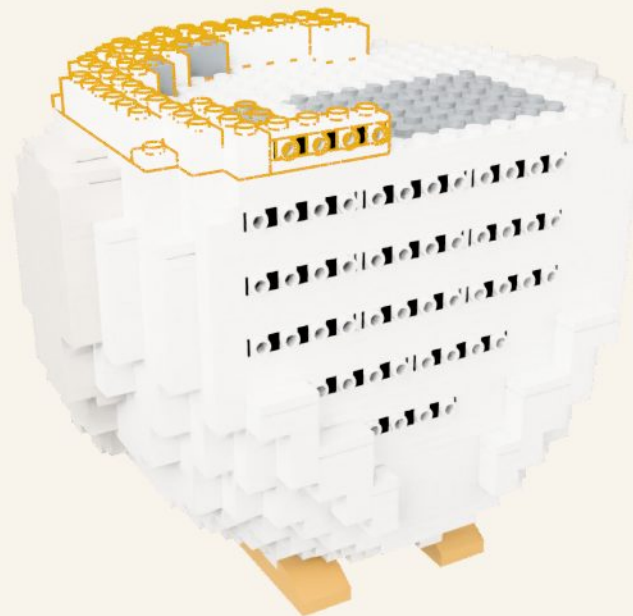
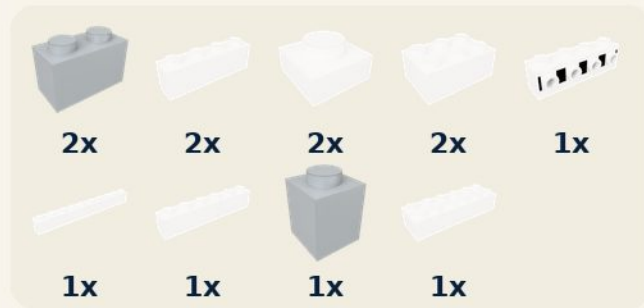
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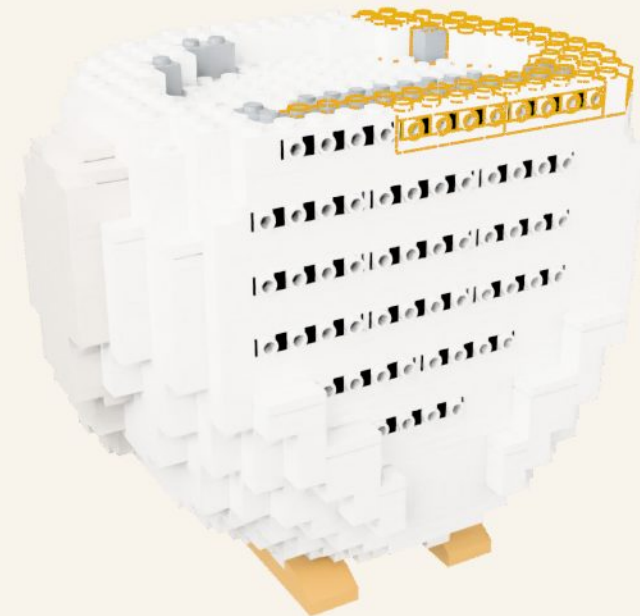
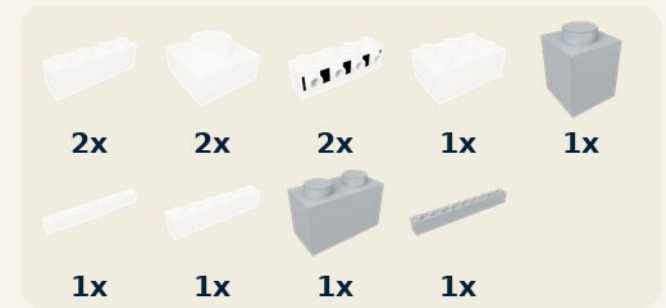
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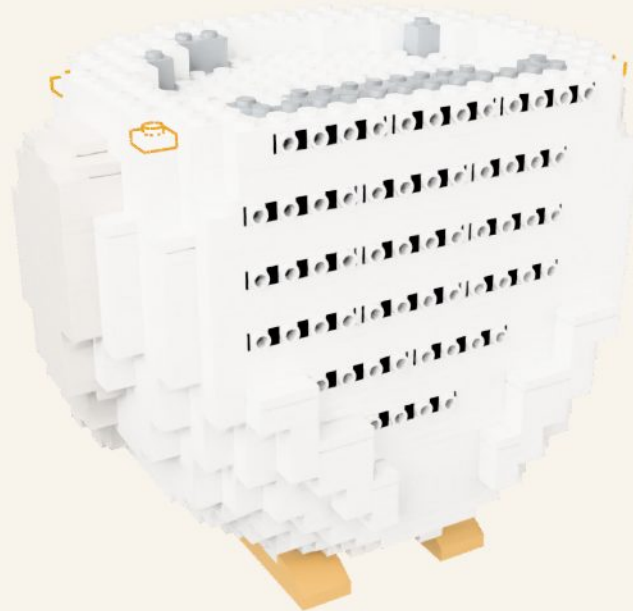
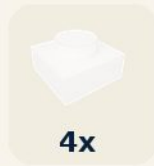
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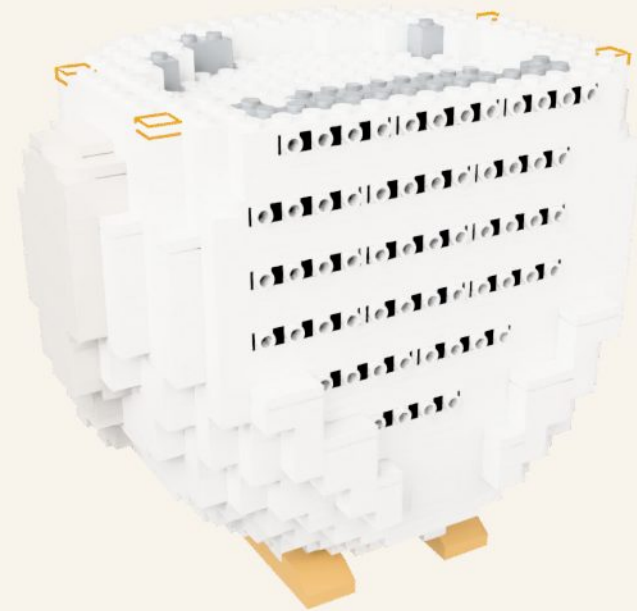
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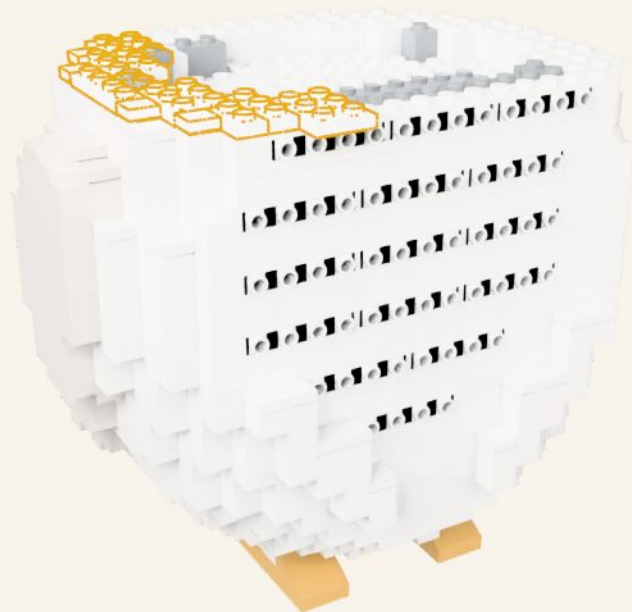
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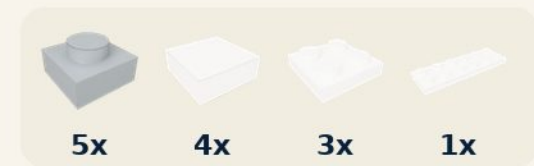
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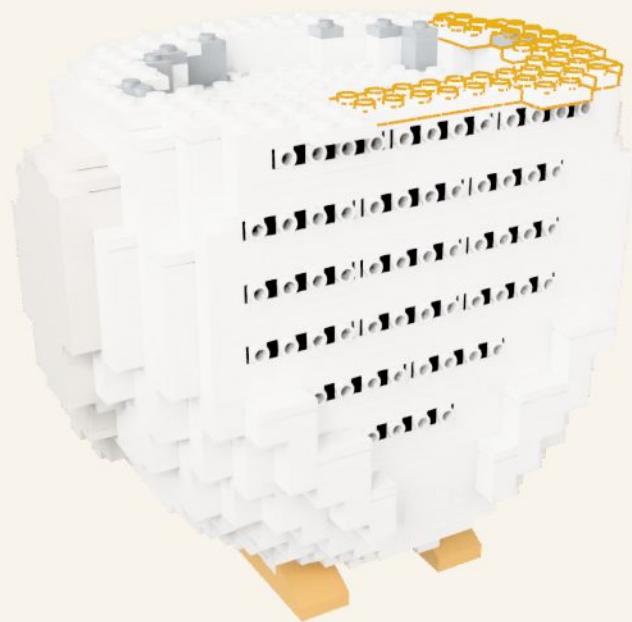
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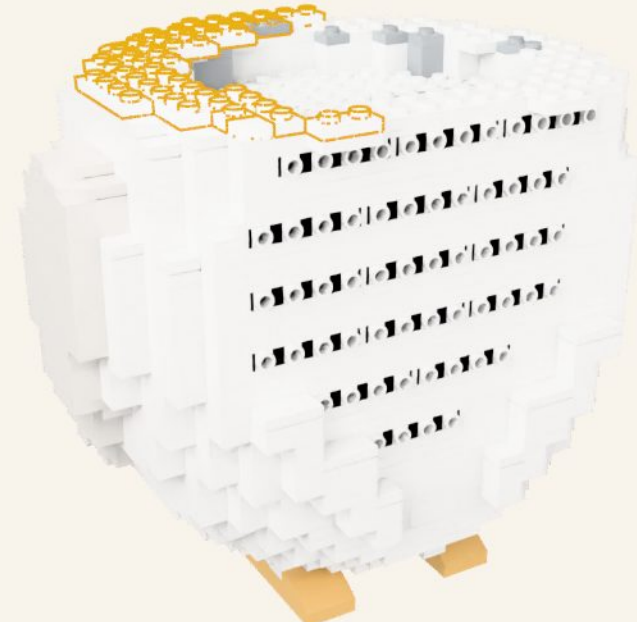
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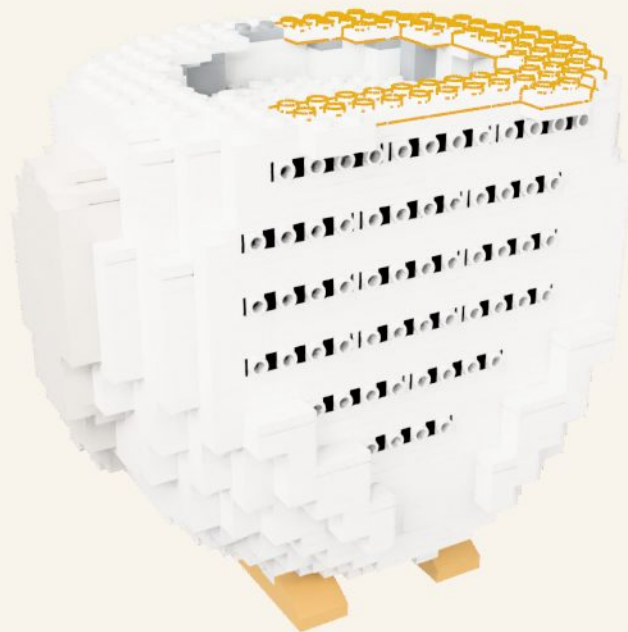
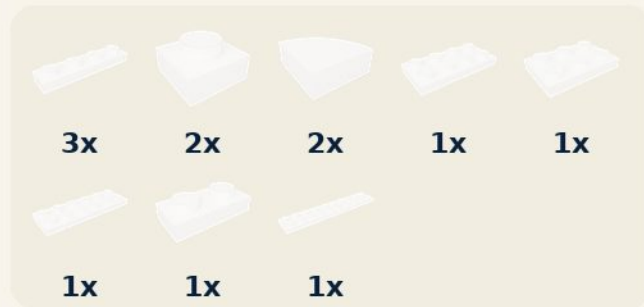
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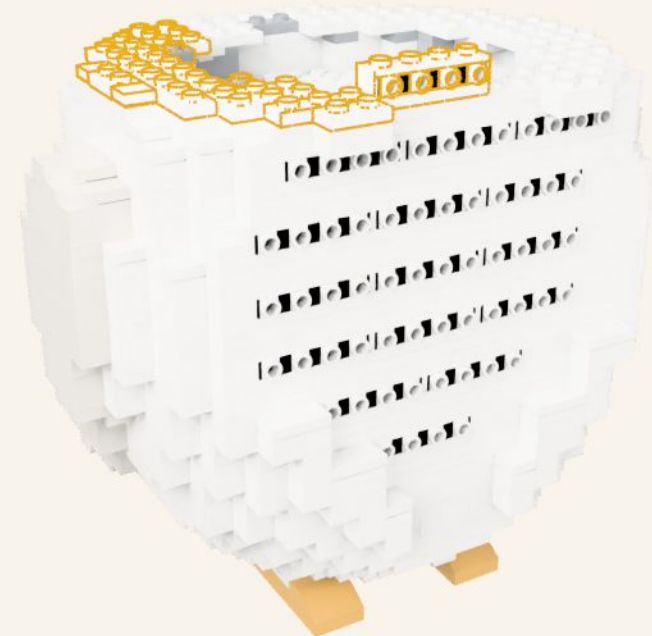
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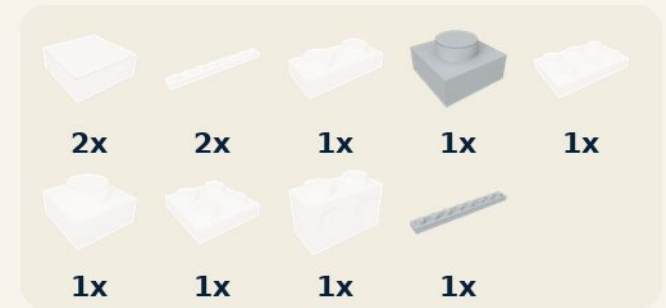
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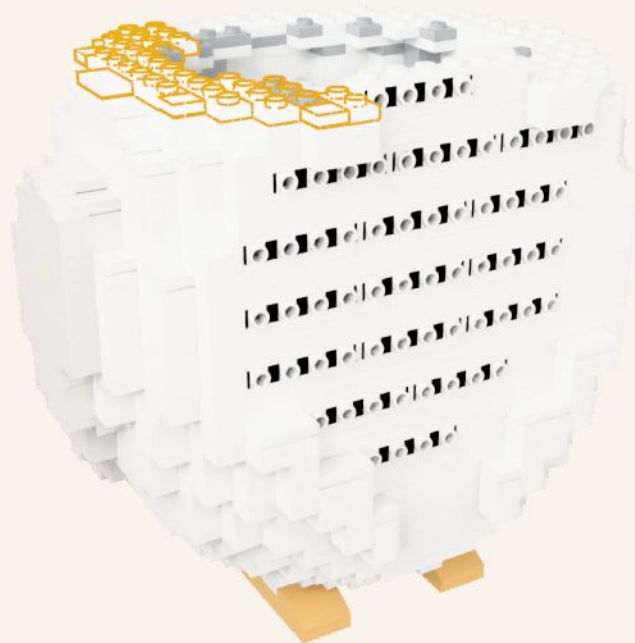
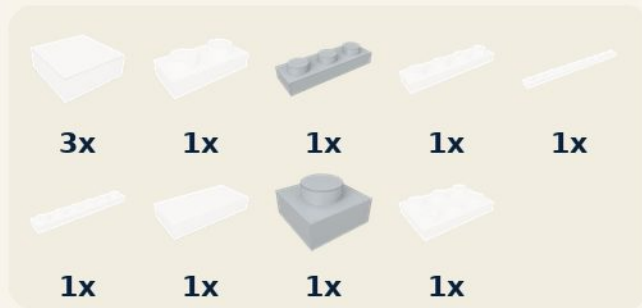
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99



100



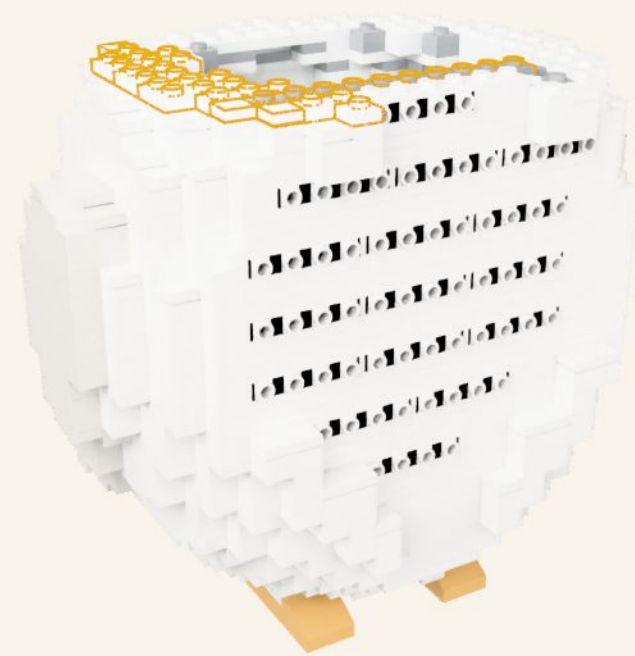
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102



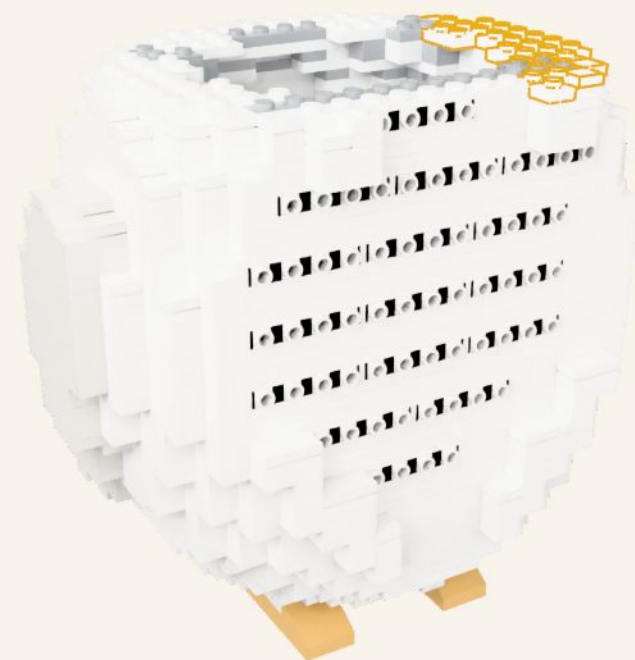
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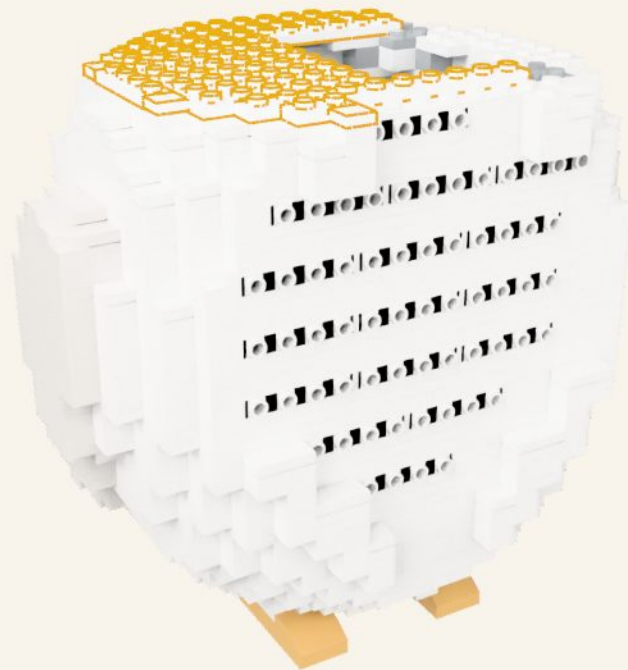
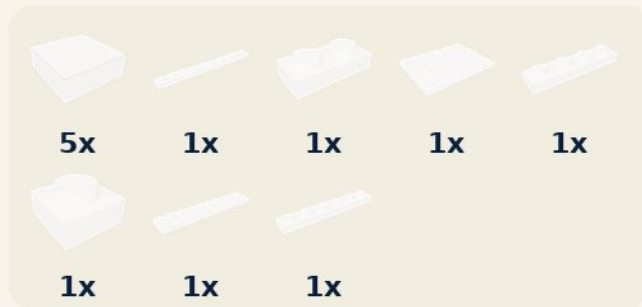
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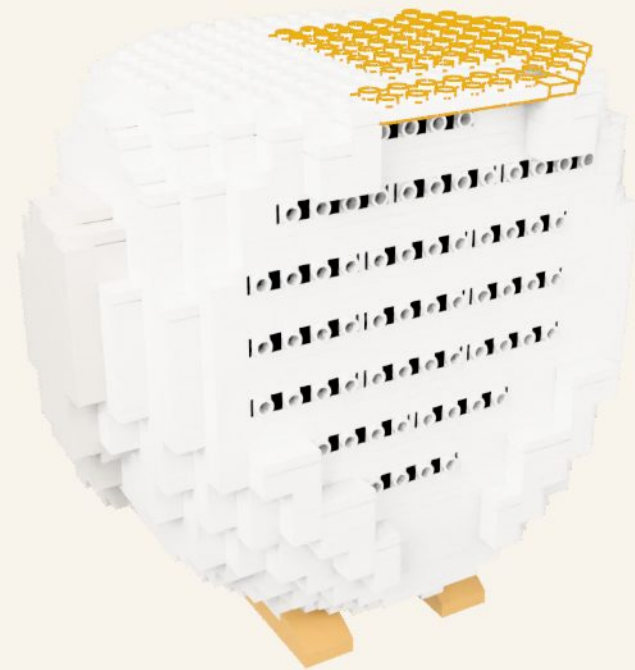
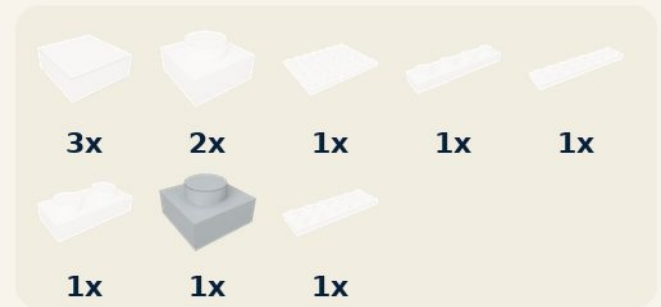
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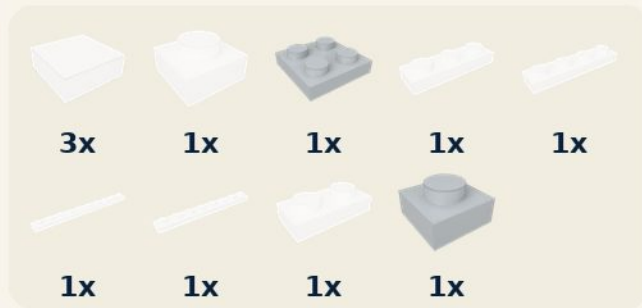
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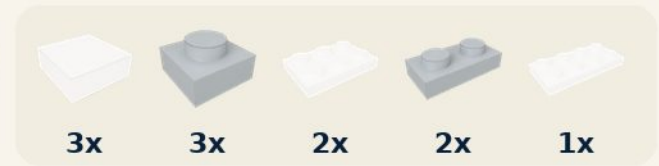
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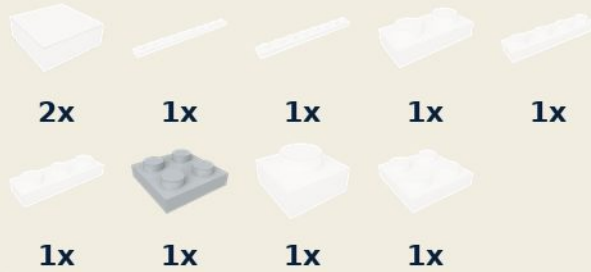
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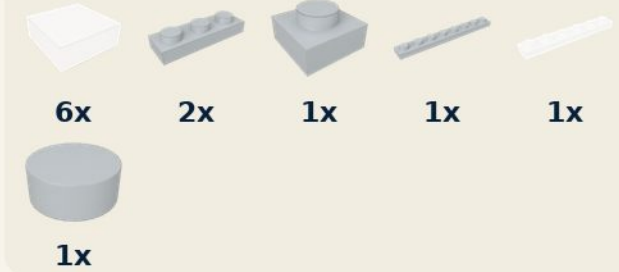
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110



111



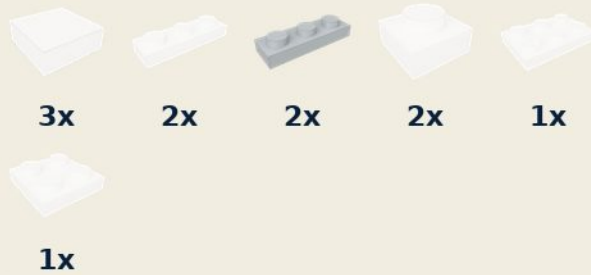
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113



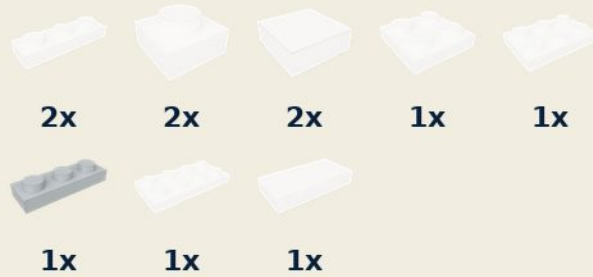
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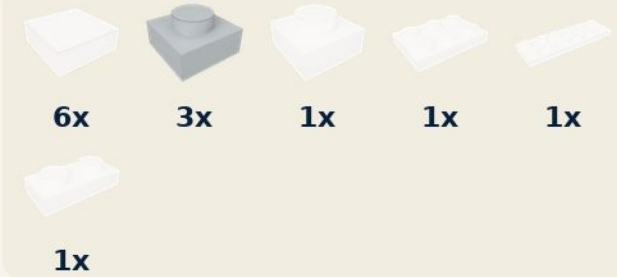
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116



117

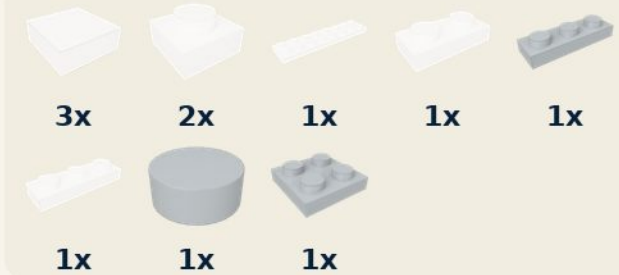


118**119**

120

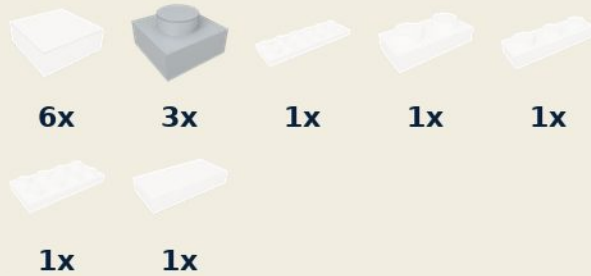


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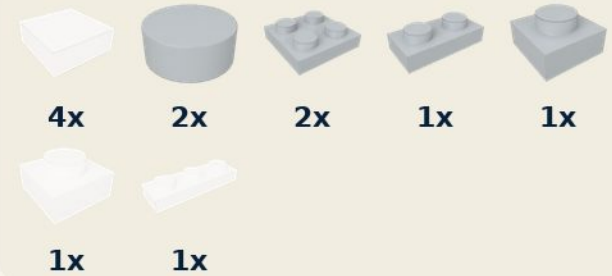


122**123**

124



125



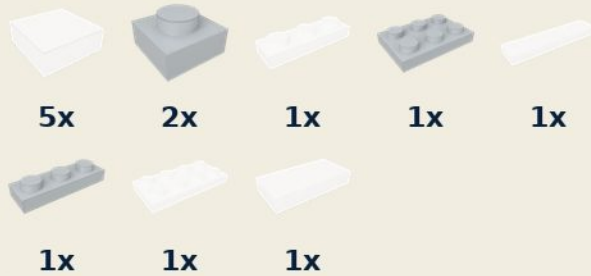
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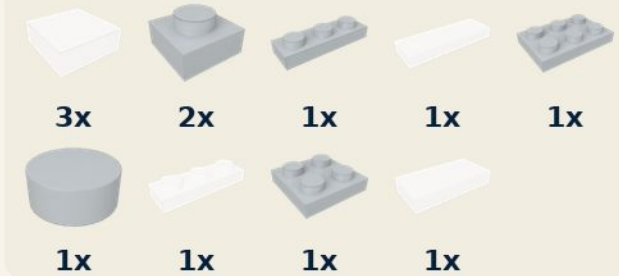
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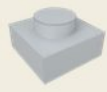


128



129



130

2x



2x



2x



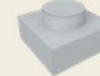
1x



1x

**131**

2x



2x



2x



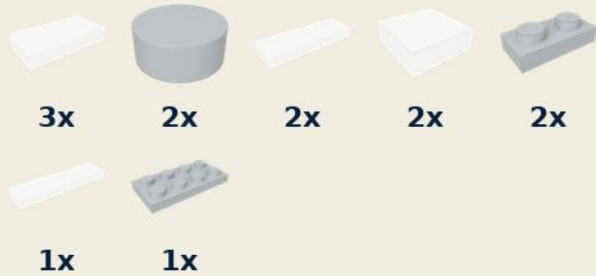
1x



1x



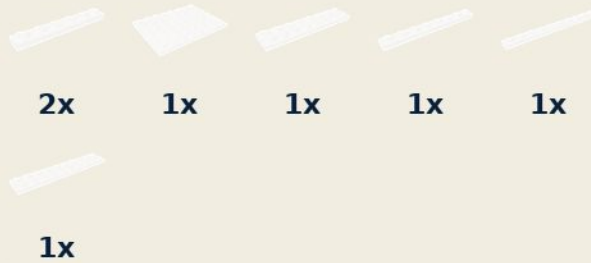
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133

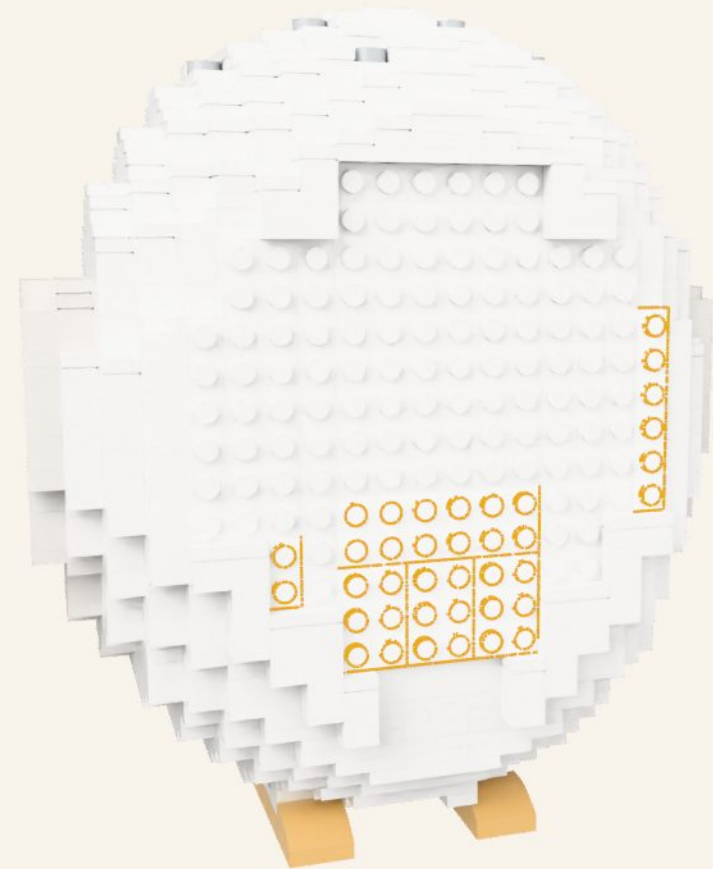


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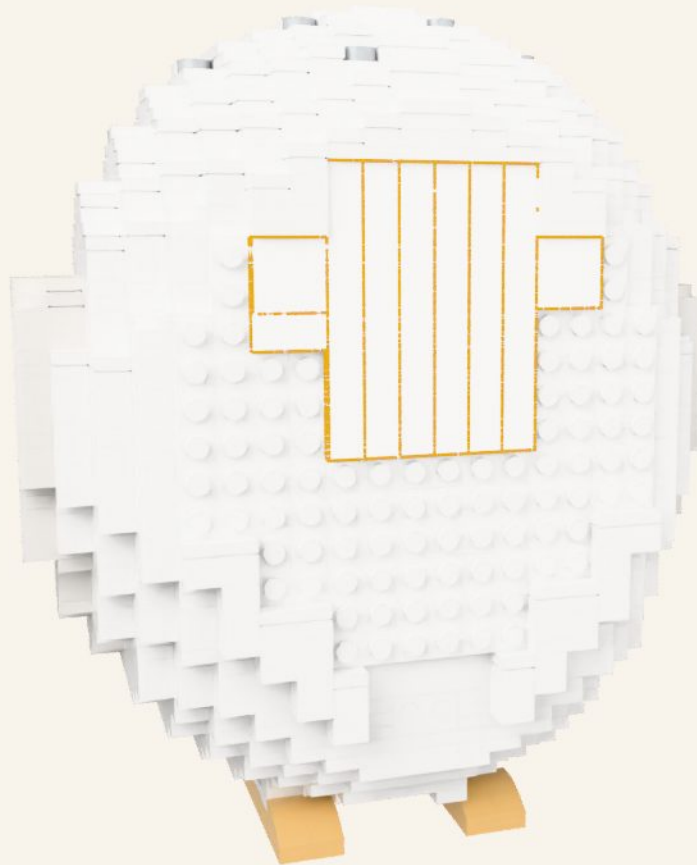
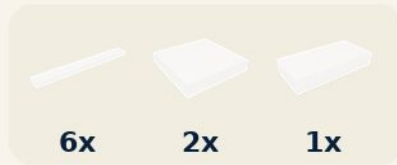


Studs forward: the face is built sideways

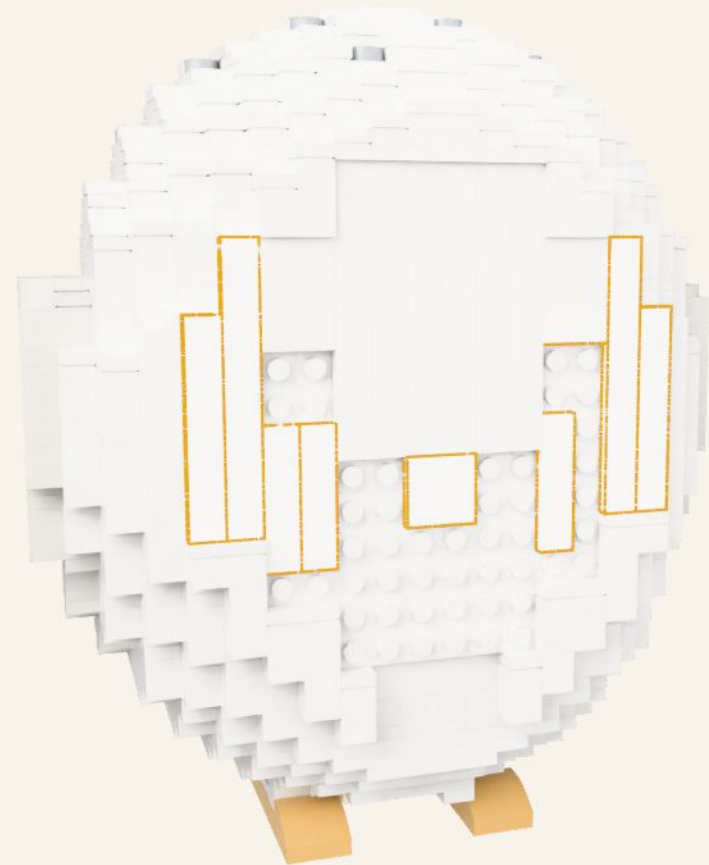
135



136



137



138



4x



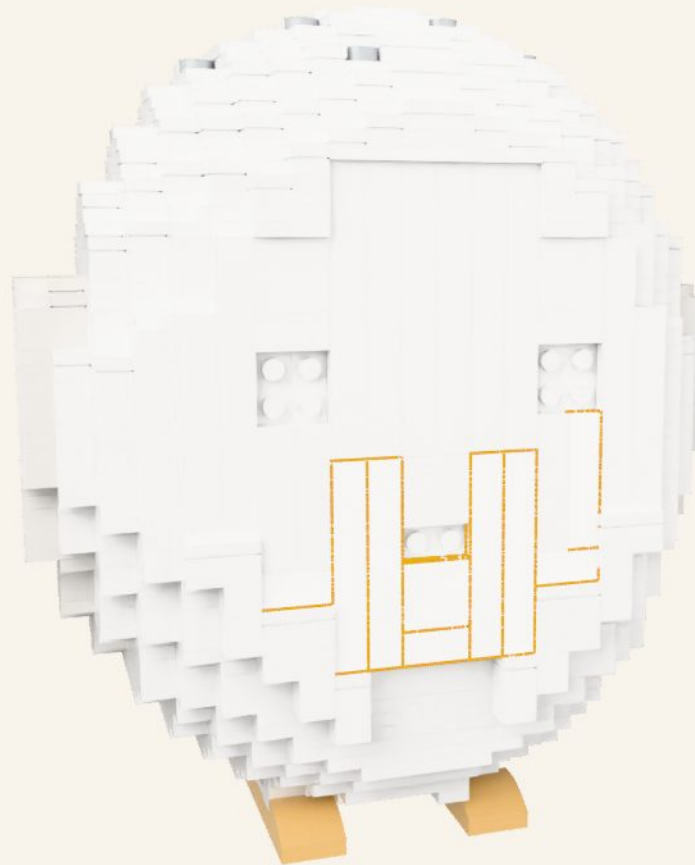
3x



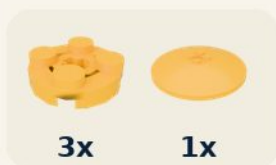
1x



1x

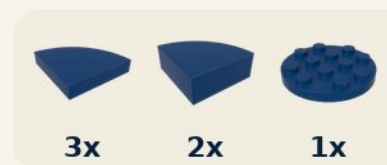


139



The left (larger) eye: iris

140

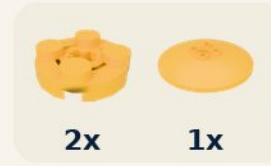


141



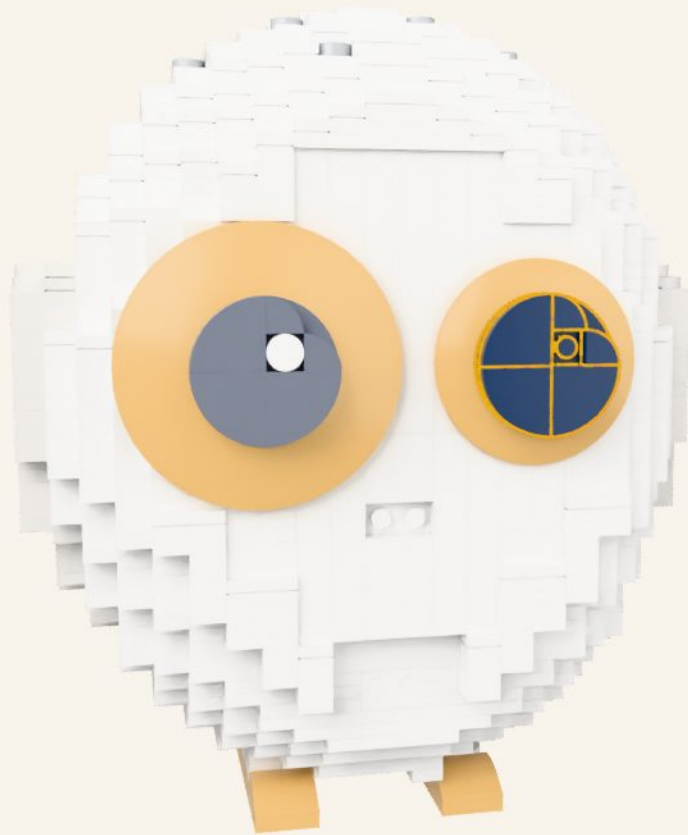
One highlight, upper right

142



The right eye: iris

143

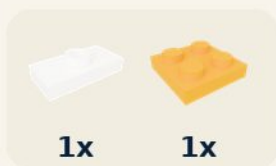


144



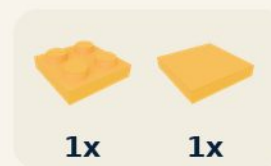
One highlight, upper right

145

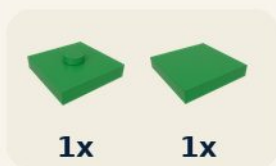


Turn the square 45 degrees on the single centre stud

146



147



Two squares, 45 degrees apart: the eight-point star

148



One soft tuft

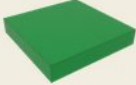
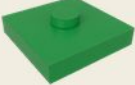
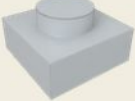
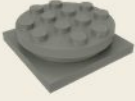
Parts inventory

1,446 pieces | 94 lots

 7x 11211 White	 13x 2431 White	 5x 2445 White	 4x 2456 White	 8x 25269 White	 5x 3001 White	 19x 3002 White
 2x 3003 White	 11x 3004 White	 23x 3005 White	 1x 3008 White	 9x 3009 White	 15x 3010 White	 25x 3020 White
 80x 3021 White	 17x 3022 White	 102x 3023 White	 133x 3024 White	 3x 3029 White	 2x 3031 White	 1x 3032 White
 13x 3034 White	 8x 3036 White	 16x 30414 White	 9x 3068b White	 32x 3069b White	 160x 3070b White	 11x 3460 White
 9x 3622 White	 58x 3623 White	 16x 3666 White	 52x 3710 White	 1x 3794b White	 31x 3795 White	 8x 3832 White

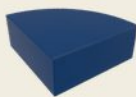
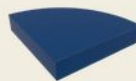
Parts inventory

1,446 pieces | 94 lots

						
12x White 4162	11x White 4477	1x White 6091	5x White 6111	3x White 63864	8x White 6636	2x White 98138
						
1x Dark Green 3068b	1x Dark Green 87580	1x Medium Stone Grey 2445	29x Medium Stone Grey 3004	27x Medium Stone Grey 3005	1x Medium Stone Grey 3009	9x Medium Stone Grey 3010
						
4x Medium Stone Grey 3020	4x Medium Stone Grey 3021	13x Medium Stone Grey 3022	56x Medium Stone Grey 3023	88x Medium Stone Grey 3024	1x Medium Stone Grey 3031	3x Medium Stone Grey 3036
						
6x Medium Stone Grey 3460	7x Medium Stone Grey 3622	23x Medium Stone Grey 3623	4x Medium Stone Grey 3666	21x Medium Stone Grey 3710	1x Medium Stone Grey 4477	5x Medium Stone Grey 6111
						
7x Medium Stone Grey 98138	1x Dark Stone Grey 3403c01	10x Light Stone Grey 25269	6x Light Stone Grey 3001	3x Light Stone Grey 3002	4x Light Stone Grey 3003	7x Light Stone Grey 3004

Parts inventory

1,446 pieces | 94 lots

 8x 3005 Light Stone Grey	 3x 3010 Light Stone Grey	 7x 3020 Light Stone Grey	 3x 3021 Light Stone Grey	 9x 3022 Light Stone Grey	 35x 3023 Light Stone Grey	 22x 3024 Light Stone Grey
 8x 3070b Light Stone Grey	 4x 3622 Light Stone Grey	 8x 3623 Light Stone Grey	 5x 3666 Light Stone Grey	 8x 3710 Light Stone Grey	 2x 63864 Light Stone Grey	 4x 11477 Flame Yellowish Orange
 2x 3022 Flame Yellowish Orange	 2x 3034 Flame Yellowish Orange	 1x 3068b Flame Yellowish Orange	 2x 3795 Flame Yellowish Orange	 1x 3961 Flame Yellowish Orange	 5x 4032b Flame Yellowish Orange	 1x 44375a Flame Yellowish Orange
 4x 25269 Earth Blue	 6x 27925 Earth Blue	 2x 60474 Earth Blue				

Engineered, then checked

- ✓ No part intersects another
- ✓ Every part is held by studs
- ✓ One connected structure
- ✓ Buildable in this order
- ✓ Comes apart again
- ✓ Stands on its feet (tips only past 16.7 deg)
- ✓ Upper body turns +/-20 deg freely

1,446 pieces | 1.5 kg | 176 x 217 x 142 mm

Checks computed from the digital model (tools/validate.py).